

NOISETTE: Certifying Differential Privacy Mechanisms Efficiently

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Abstract

Differential privacy (DP) has emerged as a rigorous framework for privacy-preserving data analysis, with widespread deployment in industry and government. Yet existing implementations typically assume that the party applying the mechanism can be trusted to sample noise correctly. This trust assumption is overly optimistic: a malicious party may deviate from the protocol to gain accuracy or avoid scrutiny, thereby undermining users' privacy guarantees.

In this paper, we introduce NOISETTE, a family of efficient protocols for certifying DP noise sampling across both discrete and continuous settings. We design a protocol that supports any discrete distribution through certifiable lookup table evaluation, and introduce a staircase-based optimization that greatly improves efficiency without compromising privacy or utility. We further extend this framework to continuous mechanisms, providing the first efficient protocol for certifiable continuous noise sampling.

We demonstrate the practicality of our protocols through concrete DP applications, including mean estimation and federated learning. Our protocols outperform the prior state-of-the-art by up to $64\times$ in runtime and $24\times$ in communication, while preserving the same accuracy as uncertified DP mechanisms. These results establish NOISETTE as the first efficient, scalable, and general-purpose solution for certifiable DP noise sampling, making certified privacy guarantees practical in high-stakes applications.

1 Introduction

Differential privacy (DP) [DMNS06a] is a rigorous mathematical framework that enables privacy-preserving data analytics and machine learning, offering strong and quantifiable privacy guarantees. For instance, Apple [DPT, DPT17] employs DP to collect usage statistics such as popular emojis and frequently visited Safari domains without revealing individual user activity. Google [goo] incorporates DP in tools like RAPPOR and DP-Fed, enabling privacy-preserving telemetry in Chrome and safeguarding user data in federated learning systems. Even government agencies have acknowledged the shortcomings of traditional anonymization methods: the U.S. Census Bureau [MKA⁺08, Abo16, otC21] recently adopted DP to protect individual records in the 2020 Census.

Many existing deployments of DP adopt the centralized model, where a trusted aggregator collects sensitive data from individuals, aggregates the results, then applies a DP mechanism that carefully adds calibrated noise before releasing the final aggregate statistics. This setup has been used in many real-world applications we mentioned above, such as technology companies reporting population-level usage statistics or agencies publishing census data.

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However, in these scenarios, the aggregator is solely responsible for correctly adding noise, and the privacy guarantees for the users are only valid if the aggregator faithfully follows the protocol. In practice, this assumption of trust can be overly optimistic. Since DP mechanisms introduce randomness to protect individual data, there is a fundamental trade-off between privacy and utility. An aggregator who prefers higher accuracy in the released results has a strong incentive to add less noise, thereby reducing privacy guarantees for users while still claiming DP protection.

Concerns about the verifiability of DP guarantees have hindered its broader adoption, particularly in settings where the outcomes influence high-stakes decisions such as resource allocation, policy evaluation, or fairness audits [LPDV23, YLBC25]. Ensuring that DP noise is both present and correctly calibrated is therefore a critical prerequisite for trustworthy DP in practice.

In response to the risk of such misuse, major technology companies are increasingly required to verify compliance with privacy requirements through independent audits. For example, as part of a discrimination settlement, Meta must report platform statistics to third-party privacy auditors [met23]. Microsoft subjects its cloud services (e.g., Azure, Office 365) to regular third-party audits to demonstrate compliance with GDPR and global privacy frameworks [mic23]. Similarly, under the Federal Trade Commission’s 2011 Google Buzz consent order, Google must maintain a comprehensive privacy program and undergo independent, biennial assessments for 20 years [ftc11]. These privacy auditing challenges highlight the critical need to certify the correct execution of DP mechanisms without revealing either the underlying data or the randomness used in the mechanism. Such certification enables privacy auditing while ensuring that any bias or utility degradation is solely attributable to the DP mechanism itself. While third-party audits are already common practice, they typically verify high-level procedural compliance or static security configurations and cannot certify each execution of a DP mechanism. This gap motivates the need for more fine-grained, privacy-preserving certification techniques that can verify correct noise generation without exposing the underlying data or randomness.

Existing work has explored addressing these challenges by certifying DP noise sampling in the context of zero-knowledge proofs (ZKPs) [BGKW24, BAT24, STC⁺24, BC23] or by designing oblivious algorithms for DP noise sampling [CsU19, WYF⁺23], which can be naturally adapted to the ZKP setting. However, these approaches face fundamental limitations in efficiency and generalizability. To address the efficiency challenge, some solutions design specialized, crypto-friendly protocols for DP noise mechanisms (e.g., Binomial), but such mechanisms often provide worse utility-privacy tradeoffs compared to the more widely used ones, such as Gaussian and Laplace [BGKW24, BC23, BAT24]. Others support (discrete or floating-point) Gaussian sampling but require expensive cryptographic computation as a result of evaluating crypto-unfriendly functions like logarithmic functions and square root [STC⁺24], or relying on rejection sampling [WYF⁺23], which is inefficient in ZKP. In addition, all existing approaches are tailored to specific DP mechanisms and cannot be easily generalized to others. Yet in practice, new DP mechanisms continue to be developed for specialized applications, such as the Cauchy mechanism for instance-based noise under smooth sensitivity [NRS07], and the Skellam mechanism for federated learning [AKL21]. Manually optimizing specialized solutions for every noise distribution is tedious and unscalable.

To address these challenges, we propose NOISETTE, a family of efficient and scalable ZKP protocols that generalize to any DP noise distribution. In summary, we make the following contributions:

Generalized Certifiable Discrete DP Noising Sampling. Our first observation is that it is possible to sample from *any discrete probability distribution* using inverse transform sampling, which represents the distribution using a finite-sized cumulative distribution function (CDF) table that is indistinguishable from the original distribution with negligible error. This technique is general and applies even to complex DP noise distributions, such as discrete Gaussian and Laplace.

Functionality $\mathcal{F}_{\text{ZK}}$

Commit: Upon receiving (commit, w) from $\mathcal{P}$ and (commit) from $\mathcal{V}$. Pick a fresh id , send (id) to both parties and store (id, w) .

Prove: Upon receiving $(\text{prove}, \mathcal{C}, (\text{id}_1, \text{id}_2, \dots, \text{id}_n))$ from $\mathcal{P}$ and $\mathcal{V}$ where $\mathcal{C} : \{0, 1\}^n \rightarrow \{0, 1\}$, retrieve values w for all tags in $\text{id} = (\text{id}_1, \text{id}_2, \dots, \text{id}_n)$. The functionality aborts if the circuits $(\mathcal{C})$ from both parties are different or any tag is invalid or if $\mathcal{C}(w) \neq 1$, otherwise sends (true) to both parties.

Plookup: Upon receiving $(\text{plookup}, a, \text{id})$ from both $\mathcal{P}$ and $\mathcal{V}$, retrieve values w for all tags in id . The functionality aborts if a is inconsistent or any tag is invalid and sends (true) to both parties if $w \subseteq a$.

Figure 1: ZK functionality supporting commit and prove.

Crucially, we make this approach highly efficient in ZKP by leveraging well-studied set inclusion protocols [PFM⁺22, ZBK⁺22, CFG24], thereby avoiding the need to design and execute deep circuits for complex noise sampling. Based on these insights, we design NOISSETTE-BASE, a generalized ZKP protocol that can efficiently sample any additive discrete DP noise.

Extension to Continuous DP Noise Distributions. Many real-world applications, however, are implemented using floating-point and require continuous (floating-point) DP mechanisms such as those used in machine learning model training [ACG⁺16a, JUO20]. Thus, it is essential to design a protocol that directly supports continuous (floating-point) DP distributions. While inverse transform sampling provides a strong starting point, representing a continuous distribution would require an astronomically large CDF table (2^{32} for FP32), making lookup operations infeasible. To overcome this challenge, we introduce a staircase approximation that groups adjacent elements into buckets. Instead of sampling across the entire domain, we reduce the complexity of the sampling by breaking it into two steps: first, sample the bucket index, and then sample within the bucket. This enables us to propose NOISSETTE-CONT, which makes sampling from continuous DP distributions feasible. As a key optimization, we also apply this staircase technique to discrete cases to construct NOISSETTE-SC, which substantially reduces the CDF table size and improves the discrete sampling efficiency. We rigorously prove that these staircase approximations preserve DP guarantees.

Evaluation on Concrete Applications. We benchmark NOISSETTE-BASE against state-of-the-art (SOTA) protocols for certifiable discrete Gaussian and Laplace sampling. Our results show that NOISSETTE-BASE outperforms SOTA protocols with up to $20\times$ faster runtime and $30\times$ lower communication overhead. We also evaluate NOISSETTE-SC and NOISSETTE-CONT in two concrete DP applications: privacy-preserving mean estimation and differentially private federated learning. Compared with SOTA protocols for continuous (floating-point) and discrete noise sampling, NOISSETTE-SC and NOISSETTE-CONT achieve up to $64\times$ speedup in runtime and $24\times$ reduction in communication, while providing the same DP and utility guarantees as uncertified mechanisms. Together, NOISSETTE demonstrates efficiency and generalizability, supporting both discrete and continuous certifiable DP noise sampling without requiring deep circuits for complex sampling.

2 Background

Notation. We use λ to denote the statistical security parameter. For a distribution $\mathcal{D}$, $x \leftarrow \mathcal{D}$ denotes x being sampled according to the distribution $\mathcal{D}$. We use upper-case letters like X (or bold

lower case letters like $\mathbf{x}$) to denote a column vector. For a vector X , X_i denotes the i -th component of X , where X_0 is the first component. We use $\llbracket x \rrbracket$ to denote the commitment of x . We use the shorthand $\llbracket \mathbf{x} \rrbracket$ to denote the commitments $\llbracket \mathbf{x}_0 \rrbracket, \llbracket \mathbf{x}_1 \rrbracket, \dots, \llbracket \mathbf{x}_n \rrbracket$. Concatenation is denoted by $\|$.

Zero-Knowledge Proof. Zero-knowledge proofs (ZKPs) for circuit satisfiability allow a prover $\mathcal{P}$ with witness w to prove to the verifier $\mathcal{V}$ that $\mathcal{C}(w) = 1$ for a public circuit $\mathcal{C}$ without revealing any information about w . We describe the functionality in Fig. 1. This functionality can be instantiated efficiently using ZKP protocols, including zkSNARKs [Gro16, CHM⁺20, GWC19] and interactive protocols [YSWW21, WYY⁺22, BMRS21, BBMH⁺21, DILO22, BBMHS22]. In this work, we use the VOLE-based commitment scheme from QuickSilver [YSWW21] as implemented in EMP-toolkit [WMK16].

Set Inclusion ZKP. Set inclusion protocols [PFM⁺22, ZBK⁺22, CFG24] allow the prover $\mathcal{P}$ to prove that $B \subset A$ where B is committed, and then the inclusion is proved based on the committed input $\llbracket B \rrbracket$ and auxiliary inputs. We describe the *plookup* functionality in Fig. 1 and the protocol from [PFM⁺22] in Fig. 16 in the appendix. First, the prover commits to an auxiliary input $C = (A\|B)$ and is sorted according to A , i.e., that either $C[i] = C[i+1]$ or if $C[i] = A[j]$ then $C[i+1] = A[j+1]$. Then, we define polynomials f and g :

$$\begin{aligned} f(\beta, \gamma) &:= (1 + \beta)^N \prod_{i \in [N]} (\gamma + B[i]) \\ &\quad \cdot \prod_{i \in [M-1]} (\gamma(1 + \beta) + A[i] + \beta A[i+1]) \\ g(\beta, \gamma) &:= \prod_{i \in [N+M-1]} (\gamma(1 + \beta) + C[i] + \beta C[i+1]) \end{aligned}$$

Both polynomials are equal iff $B \subset A$ and C is defined correctly. Both parties $\mathcal{P}$ and $\mathcal{V}$ evaluate the polynomials f and g on a random point $a, b \in \mathbb{F}_{ZK}$ and check if $f(a, b) \stackrel{?}{=} g(a, b)$. This requires $|A| + 2|B| - 2$ multiplications in $\mathbb{F}_{ZK}$.

Differential Privacy. Differential privacy (DP) [DMNS06a] mechanisms protect the privacy of sensitive data by randomizing the outputs of the query functions. The amount of privacy leakage is usually quantified by the parameters ε and δ , where ε is the privacy budget and δ is the failure probability of the DP mechanism. When $\delta = 0$, mechanism $\mathcal{M}$ is said to satisfy pure ε -DP and it satisfies approximate (ε, δ) -DP for nonzero δ . (ε, δ) -DP is a relaxation where ε -DP holds with probability at least $1 - \delta$. Smaller ε and δ indicate less privacy budget and smaller failure probability, thus stronger privacy guarantees. Below, we present the formal DP definition.

Definition 1 ((ε, δ) -DP). *A randomized mechanism $\mathcal{M}$ satisfies (ε, δ) -DP, if for all neighboring datasets $\mathcal{X}, \mathcal{X}'$ that differ in a single record, and for all possible output S :*

$$\Pr[\mathcal{M}(\mathcal{X}) \in S] \leq e^\varepsilon \cdot \Pr[\mathcal{M}(\mathcal{X}') \in S] + \delta.$$

Additive noise DP mechanisms are the most widely used DP mechanisms in practice, where additive noise is typically drawn from certain distributions like Laplace and Gaussian. For example, the Laplace mechanism $Lap(\Delta/\varepsilon)$ adds noise proportional to the function sensitivity Δ , and satisfies pure ε -DP. The function sensitivity Δ is defined as the maximum change in the output when any single record is added, removed, or replaced in the dataset.

Definition 2 (Sensitivity). *Let $\mathcal{C}_D$ be a collection of all datasets, and $f : \mathcal{C}_D \rightarrow \mathbb{R}$. The sensitivity of function f is defined as:*

$$\Delta = \max_{\mathcal{X}, \mathcal{X}' \in \mathcal{C}_D} |f(\mathcal{X}) - f(\mathcal{X}')|,$$

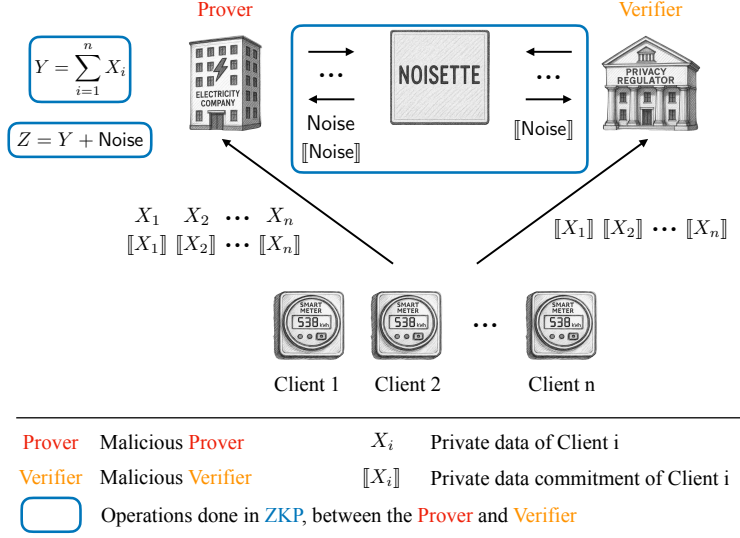


Figure 2: Setup Overview.

where the maximum is over all pairs of datasets $\mathcal{X}, \mathcal{X}' \in \mathcal{C}_D$ that differ in a single record. For higher dimensional function outputs, the difference is usually measured by the outputs' L_p norm.

There are a lot of widely used DP mechanisms for both continuous and discrete applications, including Gaussian [DR⁺14], Laplace [DR⁺14], discrete Gaussian [CKS20], discrete Laplace [GRS08], Skellam [AKL21], etc. We present detailed introductions of these mechanisms in Appendix A. In this work, we propose a generic protocol for certifying arbitrary discrete and continuous (floating-point) DP noise sampling.

In this work, we use a commonly used technique Concentrated DP (CDP) (Definition 3 from [CKS20]) to track the privacy loss, which provides a tighter and more compositional analysis than the standard (ϵ, δ) -DP framework. We can also translate CDP into concrete (ϵ, δ) -DP guarantees. $(\epsilon, 0)$ -DP implies $\frac{1}{2}\epsilon^2$ -CDP and $\frac{1}{2}\epsilon^2$ -CDP implies $(\frac{1}{2}\epsilon^2 + \epsilon\sqrt{2\log(1/\delta)}, \delta)$ -DP, $\forall \delta > 0$.

Definition 3 (CDP). A randomized algorithm $\mathcal{M} : \mathcal{X}^n \rightarrow \mathcal{Y}$ satisfies $\frac{1}{2}\epsilon^2$ -CDP if, $\forall x, x' \in \mathcal{X}^n$ differing on a single element and $\forall \alpha > 1$, we have $D_\alpha(\mathcal{M}(x) || \mathcal{M}(x')) \leq \frac{1}{2}\epsilon^2\alpha$, where $D_\alpha(P || Q) = \frac{1}{\alpha-1} \log(\sum_{y \in \mathcal{Y}} P(y)^\alpha Q(y)^{1-\alpha})$ is the Rényi divergence of order α of distribution P from distribution Q .

DP is widely adopted in machine learning model training, and **DP-SGD** [ACG⁺16b] is a widely used optimization technique for differentially private model training. DP-SGD provides a DP guarantee for the trained model by clipping the model gradients during training and then applying a DP mechanism, such as Gaussian mechanism.

3 Overview

Before introducing our techniques, we first outline the setup of DP certification with a concrete example. Consider an electricity company that collects data from household smart meters. These measurements are valuable for demand forecasting and grid optimization, but they also reveal sensitive living patterns, such as when residents are home or asleep. To protect privacy, the company can apply DP by adding calibrated noise before releasing aggregate statistics. The guarantee,

however, depends entirely on the aggregator applying the DP mechanism correctly. The aggregator may have incentives to reduce noise for higher accuracy or add excessive noise to avoid scrutiny, while it cannot publish raw data due to public and regulatory pressure. In our setting, each smart meter (client) commits to its collected data and forwards the commitments to both the aggregator and an independent regulator. The aggregator acts as the prover, generating DP-protected statistics, and the regulator is the verifier that ensures they were produced with the promised DP mechanism. We present the setup in Fig. 2 to ease understanding our setting.

Threat Model. We consider a set of clients holding sensitive data and a central server that aggregates the results. The party responsible for adding DP noise (whether it is the central server or the clients themselves) acts as the prover. We consider prover to be malicious and may deviate from the DP mechanism in order to gain higher accuracy or infer other clients data if clients are responsible for adding DP noise, but it will not release raw data due to public and regulatory pressure. An independent regulator acts as the verifier, checking that the reported outputs were produced with the promised DP mechanism. The verifier can also be malicious and may be curious about the prover’s private data.

We assume clients will honestly commit their data, meaning that they do not trust other parties with their sensitive information but will honestly follow the protocol for data commitment and transfer. We do not consider data manipulation attacks such as backdoor attacks in federated learning. Depending on the application, the clients may either (1) delegate noise addition to the server aggregator, or (2) add DP noise themselves and prove correctness to the verifier. In both cases, the prover may deviate from the protocol, while the verifier ensures correctness without collusion. We assume no collusion among clients, the prover, and the verifier.

The essential task in this setting is DP noise sampling certification, which NOISSETTE is designed to support efficiently. For discrete DP mechanisms, we employ NOISSETTE-BASE and its staircase-optimized variant NOISSETTE-SC, as described in § 4 and § 5. When the data is continuous and represented using floating-point, a continuous DP mechanism is required, and we use NOISSETTE-CONT for efficient continuous (floating-point) noise sampling, also introduced in § 5. We further discuss precision issues in continuous noise representations in § 5.3, and demonstrate the efficiency of our protocols through concrete DP applications in § 6.

4 Certifying Discrete Noise Sampling

In this section, we introduce NOISSETTE-BASE, a new zero-knowledge protocol for certifying sampling correctness for *any* discrete DP noise distribution. Fig. 3 describes the corresponding ideal functionality $\mathcal{F}_{\text{sample}}$. $\mathcal{F}_{\text{sample}}$ takes as input two parameters: $\mathcal{D}$, a finite and discrete noise distribution; and N , number of samples. It then samples $\mathbf{x}_i \leftarrow \mathcal{D}$ for $i \in [N]$. $\mathcal{F}_{\text{sample}}$ outputs these samples $\mathbf{x}$ to prover $\mathcal{P}$, and commitments $[\![\mathbf{x}]\!]$ to both $\mathcal{P}$ and verifier $\mathcal{V}$.

To design an efficient protocol, our first observation is that, given any concentrated DP noise distribution, it is possible to truncate to a finite domain such that the truncated distribution is statistically indistinguishable from the original distribution. The truncated distribution can be easily instantiated using a lookup table that encodes its CDF. We then make use of a technique called *inverse transform sampling* to sample directly from the CDF table. For the rest of this section, we first formally describe properties of truncation, followed by inverse transform sampling, how to generate uniform randomness, and finally the NOISSETTE-BASE protocol.

<u>Functionality $\mathcal{F}_{\text{sample}}$</u>	
Input:	Number of samples N , finite and discrete distribution to be sampled from $\mathcal{D}$.
Functionality:	Upon receiving $(\text{sample}, \mathcal{D}, N)$ from $\mathcal{P}$ and $\mathcal{V}$, sample $\mathbf{x}_i \leftarrow \mathcal{D}$ for $i \in [N]$.
Output:	Send $\mathbf{x}$, $\llbracket \mathbf{x} \rrbracket$ to $\mathcal{P}$ and $\llbracket \mathbf{x} \rrbracket$ to $\mathcal{V}$.

Figure 3: Functionality for sampling $\mathbf{x} \leftarrow \mathcal{D}$.

Distribution	Truncation Bound M
Discrete Gaussian $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$	$\sigma \sqrt{(2 \ln 2)(\lambda + 1) + 2 \ln N} + 1$
Continuous Gaussian $\mathcal{N}(0, \sigma^2)$	$\sigma \sqrt{(2 \ln 2)(\lambda + 1) + 2 \ln N}$
Discrete Laplace $Lap_{\mathbb{Z}}(0, t)$	$t \ln \frac{2N \cdot 2^\lambda}{e^{1/t} + 1}$
Continuous Laplace $Lap(0, t)$	$t \ln(N \cdot 2^\lambda)$
Skellam $Sk(\mu)$	$1.0862\mu + \lambda \ln 2 + \ln N$

Table 1: Truncation parameters M for commonly used DP noise distributions. λ denotes the statistical security parameter, and N is the total number of samples to be drawn.

4.1 Distribution Truncation

Any concentrated discrete DP distribution $\mathcal{D}$ can be truncated to distribution $\mathcal{D}'$ by first limiting $\mathcal{D}$'s domain to $[-M, M]$, and then normalizing the probabilities such that they sum to 1. Lemma 1 formally defines statistical indistinguishability between $\mathcal{D}$ and $\mathcal{D}'$ in terms of the security parameter λ .

Lemma 1. *Let $\mathcal{D}$ be a discrete distribution over $\mathbb{Z}$ and suppose $\Pr_{X \sim \mathcal{D}}(|X| > M) \leq f(M)$. Let $\mathcal{D}'$ be the distribution obtained by truncating $\mathcal{D}$ at $[-M, M]$, i.e., for all $x \in [-M, M]$*

$$\Pr_{X' \sim \mathcal{D}'}(X' = x) = \frac{\Pr_{X \sim \mathcal{D}}(X = x)}{\Pr_{X \sim \mathcal{D}}(|X| \leq M)}$$

Then for any PPT adversary $\mathcal{A}$ that sees N i.i.d. samples, $\mathcal{D}$ and $\mathcal{D}'$ are statistically indistinguishable if $Nf(M) \leq 2^{-\lambda}$ for statistical security parameter λ .

Truncation Parameters. Most additive DP noise distributions are concentrated including Gaussian [CKS20, DR⁺14], Laplace [CKS20], Skellam [AKL21], etc. We use Chernoff bound [Che52] for each of these distributions to determine the truncation parameter and describe them in Tab. 1. We include both discrete and continuous distributions here and will discuss sampling from continuous distributions in § 5.3.

4.2 Inverse Transform Sampling

After truncating the discrete distribution $\mathcal{D}$ to $\mathcal{D}'$ with a finite domain, we use the inverse transform method to sample from $\mathcal{D}'$. The CDF is defined as $\text{CDF}_{\mathcal{D}'}(x) = \Pr_{X \sim \mathcal{D}'}(X \leq x)$. We use a sorted vector $\mathbf{x}$ to denote the support of $\mathcal{D}'$ such that $\text{CDF}_{\mathcal{D}'}(\mathbf{x}_{i-1}) < \text{CDF}_{\mathcal{D}'}(\mathbf{x}_i)$ and $\text{CDF}_{\mathcal{D}'}(\mathbf{x}_{-1}) := 0$. Then the probability $\Pr_{X \sim \mathcal{D}'}(X = \mathbf{x}_i) = \text{CDF}_{\mathcal{D}'}(\mathbf{x}_i) - \text{CDF}_{\mathcal{D}'}(\mathbf{x}_{i-1})$.

Thus, we represent the distribution $\mathcal{D}'$ as a table T with its i -th entry defined as $T[i] = (\text{CDF}_{\mathcal{D}'}(\mathbf{x}_{i-1}), \text{CDF}_{\mathcal{D}'}(\mathbf{x}_i), \mathbf{x}_i)$. To sample from the distribution $\mathcal{D}'$, we first sample a uniform value

Functionality $\mathcal{F}_{\text{rand}}$

This functionality has all the same features as $\mathcal{F}_{\text{zk}}$ and the following command:

Rand. Upon receiving(rand, n) from $\mathcal{P}$ and $\mathcal{V}$, sample $\mathbf{x} \leftarrow \{0, 1\}^n$ and send ($\mathbf{id}$) to both parties and store $(\mathbf{id}, \mathbf{x})$.

Figure 4: Functionality to get committed random bits.

$u \in (0, 1]$. Then, we go through the CDF table and find an index i such that $\text{CDF}_{\mathcal{D}'}(\mathbf{x}_{i-1}) < u \leq \text{CDF}_{\mathcal{D}'}(\mathbf{x}_i)$. The sampled value is $\mathbf{x}_i$.

Optimization for Symmetric Distributions. For symmetric distributions, we can halve the table size to sample the distribution’s right half and additionally sample a sign bit to extend the sampled value to the whole domain. Thus, we construct the CDF table containing $M + 1$ elements and sample $\mathbf{x}_i \in [0, M]$ using the procedure mentioned above. Then we sample a random sign indicator $s \in \{-1, 1\}$. And the output is $s \cdot \mathbf{x}_i$.

Distribution Shift under Fixed-Point Representation. Inverse transform sampling assumes real-valued CDF probabilities and uniform $u \in (0, 1]$. In practice, however, finite-precision computation cannot represent arbitrary reals. We therefore use n -bit fixed-point to encode both the CDF probabilities and the uniform value u . We show that this induces negligible impacts on privacy guarantee.

Lemma 2. *Consider a CDF table with M entries. Let δ_2 denote the probability that, over N samples, at most one output differs from the ideal real-valued inverse-CDF sample when all CDF probabilities and the uniform $u \in (0, 1]$ are represented using n -bit fixed-point numbers. This requirement is met whenever*

$$n \geq \log N + \log M - \log \delta_2$$

We describe the table construction in detail in Appendix F. Now the essential part is to sample an n -bit uniform value u ; we address how to ensure fair randomness in the next section.

4.3 Fair Randomness

To sample a uniform n -bit fixed-point u , $\mathcal{F}_{\text{sample}}$ makes use of an ideal functionality $\mathcal{F}_{\text{rand}}$ that takes the number of bits n as input and outputs n random bit commitments to both parties (see Fig. 4). We instantiate the $\mathcal{F}_{\text{rand}}$ functionality using Π_{rand} as shown in Fig. 5. The prover $\mathcal{P}$ samples n random bits and sends their commitments to the verifier $\mathcal{V}$. Upon receiving these commitments from the $\mathcal{P}$, $\mathcal{V}$ sends a random seed s to $\mathcal{P}$. $\mathcal{P}$ then uses s to generate n random bits and outputs the final bits as the XOR of both $\mathcal{P}$ ’s and $\mathcal{V}$ ’s randomness. We note that the interaction here cannot be avoided using the Fiat-Shamir transform because $\mathcal{P}$ should not know the random seed of $\mathcal{V}$ before committing to its own random bits.

4.4 Noisette-Base Protocol Design

We present our full sampling protocol NOISETTE-BASE in Fig. 6. To sample from $\mathcal{D}'$, firstly, $\mathcal{P}$ and $\mathcal{V}$ use Π_{rand} to obtain the commitment of a uniformly random value $\llbracket u \rrbracket$. $\mathcal{P}$ outputs $\mathbf{x}_i$ such that $\text{CDF}(\mathbf{x}_{i-1}) < u \leq \text{CDF}(\mathbf{x}_i)$. $\mathcal{P}$ commits to $\mathbf{x}_i$, $\text{CDF}(\mathbf{x}_{i-1})$ and $\text{CDF}(\mathbf{x}_i)$. To prove that the sampling was done honestly, the parties verify that the committed values satisfy the condition

Protocol Π_{rand}

$\mathcal{R}$ is a random oracle.

1. $\mathcal{P}$ samples $\mathbf{x} \leftarrow \{0, 1\}^n$ and sends $\llbracket \mathbf{x} \rrbracket$ to the $\mathcal{V}$.
2. On receiving $\llbracket \mathbf{x} \rrbracket$, $\mathcal{V}$ sends $s \leftarrow \{0, 1\}^\kappa$ to $\mathcal{P}$.
3. $\mathcal{P}$ and $\mathcal{V}$ obtain $\mathbf{y} \leftarrow \mathcal{R}(s)$
4. $\mathcal{P}$ receives $\mathbf{r}, \llbracket \mathbf{r} \rrbracket$ and $\mathcal{V}$ receives $\llbracket \mathbf{r} \rrbracket$, where $\mathbf{r} = \mathbf{x} \oplus \mathbf{y}$.

Figure 5: Protocol for sampling n uniformly random bits.

$\text{CDF}(\mathbf{x}_{i-1}) < u \leq \text{CDF}(\mathbf{x}_i)$. Finally, we need to prove that the committed tuple exists in the table T that defines the public distribution $\mathcal{D}'$. Both parties compute the set defined as follows:

$$A = \{\alpha^2 \mathbf{x}_i + \alpha \text{CDF}(\mathbf{x}_i) + \text{CDF}(\mathbf{x}_{i-1}) | i \in [N]\}$$

where $\alpha \leftarrow \mathbb{F}_{ZK}$ is a random value chosen by the verifier $\mathcal{V}$, and $\mathbb{F}_{ZK}$ denotes the finite field over which $\mathcal{F}_{ZK}$ defines its arithmetic constraints. $\mathcal{P}$ and $\mathcal{V}$ compute the same linear combination for the chosen value and execute a set inclusion check. When sampling N values, the set inclusion check can be amortized across all samples. We use the Plookup protocol [GW20] Π_{plookup} for set inclusion check (Fig. 16) and $\mathcal{F}_{ZK}$ to verify the range check. For clarity, Fig. 6 omits the (optional) sign-bit sampling used for symmetric noise; it can be added exactly as in § 4.2 without affecting the following analysis.

Complexity Analysis. We analyze the cost of obtaining N samples from $\mathcal{D}'$ where M is its CDF table size. The prover and verifier perform $2N + M - 2$ multiplications in $\mathbb{F}_{ZK}$ for Π_{plookup} at step 5b. For step 5a, we require 2 comparisons per sample. Thus, we need $O(2N)$ AND gates for N samples. We use the Fiat-Shamir technique to make step 3 non-interactive. The cost per noise sample is thus $O(M/N + c')$ where c' is a small constant. Thus, NOISSETTE-BASE can scale to large number of noise samplings, and the cost is amortized over the number of samples N .

Security Analysis. Let $\mathcal{D}'$ be the truncation of a discrete DP noise distribution $\mathcal{D}$ as in Lemma 1; adding noise from $\mathcal{D}$ is (ε, δ) -DP. In the $(\mathcal{F}_{ZK}, \mathcal{F}_{\text{random}})$ -hybrid model, Theorem 1 shows that Π_{sample} realizes $\mathcal{F}_{\text{sample}}(\mathcal{D}', N)$: honest execution outputs i.i.d. samples from $\mathcal{D}'$ with consistent commitments, any deviation by a malicious prover is detected except with probability at most $2N/|\mathbb{F}_{ZK}|$ (plus negligible instantiation error), and a corrupted verifier's view is simulatable by zero-knowledge. When $\mathcal{D}'$ is implemented via n -bit fixed-point CDF values, Lemma 2 bounds the additional sampling deviation by δ_2 , and Lemma 1 bounds the truncation error from $\mathcal{D}$ to $\mathcal{D}'$ by δ_1 ; thus these errors can be absorbed as additive slack in the final (ε, δ) -DP guarantee as shown in Corollary 1. We present the proof of Theorem 1 in Appendix B.

Theorem 1. *The protocol Π_{sample} securely instantiates the functionality $\mathcal{F}_{\text{sample}}$ in the $(\mathcal{F}_{ZK}, \mathcal{F}_{\text{rand}})$ -hybrid model for any finite and discrete distribution $\mathcal{D}'$.*

Corollary 1 (DP Guarantee of NOISSETTE-BASE). *Let δ_1 be the small truncation error from Lemma 1 (over N samples), and let δ_2 be the small fixed-point deviation bound from Lemma 2. Let $\delta_{\text{cert}} := \text{negl}(\kappa) + 2N/|\mathbb{F}_{ZK}|$ be the soundness error induced in Step 5 (see proof of Theorem 1*

Protocol Π_{sample}

Input:

- Truncated distribution to be sampled from $\mathcal{D}'$.
- Number of samples N .

Publicly Compute:

- $\mathcal{P}$ and $\mathcal{V}$ construct the public table $T \leftarrow \Pi_{\text{TConstruct}}(\mathcal{D}', N)$, where $T_i = (p_i^{\min}, p_i^{\max}, \mathbf{x}_i)$, $i \in [M]$, $p_i^{\min}, p_i^{\max} \in \{0, 1\}^n$ and $\mathbf{x}_i \in \mathcal{D}'$.

Protocol:

1. $\mathcal{P}$ and $\mathcal{V}$ send (rand, nN) to $\mathcal{F}_{\text{rand}}$. $\mathcal{P}$ obtains U and $\llbracket U \rrbracket$ and $\mathcal{V}$ obtains $\llbracket U \rrbracket$, where $\llbracket U \rrbracket = \{\llbracket U_0 \rrbracket, \llbracket U_1 \rrbracket, \dots, \llbracket U_{N-1} \rrbracket\}$, $U_i \in \{0, 1\}^n$.
 2. $\mathcal{P}$ defines the vector S , where $S_i := T_j$, s.t. $p_j^{\min} < U_i \leq p_j^{\max}$, $i \in [N]$. $\mathcal{P}$ and $\mathcal{V}$ use $\mathcal{F}_{\text{ZK}}$ to obtain the commitments to $S_{i,0}$, $S_{i,1}$ and $S_{i,2}$ for all $i \in [N]$.
 3. $\mathcal{V}$ samples $\alpha \leftarrow \mathbb{F}_{\text{ZK}}$ and sends it to $\mathcal{P}$.
 4. $\mathcal{P}$ and $\mathcal{V}$ compute A and $\llbracket B \rrbracket$ such that $A_j = T_{j,0} + \alpha T_{j,1} + \alpha^2 T_{j,2}$, $j \in [M]$ and $\llbracket B_i \rrbracket = \llbracket S_{i,0} \rrbracket + \alpha \llbracket S_{i,1} \rrbracket + \alpha^2 \llbracket S_{i,2} \rrbracket$, $i \in [N]$ respectively.
 5. $\mathcal{P}$ and $\mathcal{V}$ perform the following checks:
 - (a) For all $i \in [N]$, $\mathcal{P}$ and $\mathcal{V}$ use $\mathcal{F}_{\text{ZK}}$ to prove that $S_{i,0} < U_i \leq S_{i,1}$.
 - (b) $\mathcal{P}$ and $\mathcal{V}$ send $(\text{plookup}, A, \llbracket B \rrbracket)$ to $\mathcal{F}_{\text{ZK}}$ to check $B \subseteq A$.
- If both checks pass, $\mathcal{V}$ outputs true; otherwise, $\mathcal{V}$ outputs false.

Output: $\mathcal{P}$ receives $S_{i,2}$, $\llbracket S_{i,2} \rrbracket$ and $\mathcal{V}$ receives $\llbracket S_{i,2} \rrbracket$, $i \in [N]$.

Figure 6: NOISETTE-BASE protocol for verified sampling from a finite and discrete distribution $\mathcal{D}'$.

in Appendix B). If an ideal mechanism that uses N i.i.d. samples from $\mathcal{D}$ is (ε, δ) -DP, then the mechanism that obtains its N noise samples via Π_{sample} is

$$(\varepsilon, \delta + \delta_1 + \delta_2 + \delta_{\text{cert}})\text{-DP}.$$

Public verifiability. Our protocol naturally extends to the public verifiability setting described in [BGKW24]. For this adaptation, we assume the existence of a public registrar. During the setup phase, the prover $\mathcal{P}$ and the registrar jointly generate commitments to unbiased random bits using $\mathcal{F}_{\text{rand}}$. When the verifier $\mathcal{V}$ issues a query q to both the prover and the registrar, the registrar responds by sending the commitments to the randomness. Given these commitments, $\mathcal{P}$ and $\mathcal{V}$ can execute steps 3–5 to verify the sampling process.

5 Efficient Continuous Noise Sampling via Staircase Optimization

With NOISSETTE-BASE, we can efficiently sample from any discrete DP noise distribution. However, NOISSETTE-BASE cannot be directly generalized to sampling continuous (floating-point) DP noise. Naively representing floating-point DP noise using CDF table leads to a very large table – the support of the floating-point distribution can be as large as 2^{32} if FP32 is used.

To address this challenge, we propose a novel technique called *staircase approximation*. Our insight is that, by slightly modifying the original DP noise distribution to a staircase approximation, we can significantly reduce the CDF table size. Reducing the table size 1) improves NOISSETTE-BASE’s efficiency for discrete noise sampling, and 2) enables efficient continuous noise sampling. We rigorously prove staircase optimizations maintain privacy guarantees.

In the rest of this section, we first explain how to use staircase approximation to reduce a discrete noise distribution’s CDF table size, and derive a more efficient discrete noise sampling protocol NOISSETTE-SC. Then, we generalize our optimization to build a certifiable continuous noise sampling protocol NOISSETTE-CONT.

5.1 Staircase DP Noise

In this section, we first introduce the discrete staircase DP noise, and then rigorously analyze its privacy guarantees.

To construct a discrete staircase DP noise from a symmetric, truncated discrete distribution $\mathcal{D}'$, we first represent its CDF using a table of M probabilities. Our algorithm then takes a bucket size parameter, m , and reduces these to $\frac{M}{m}$ probabilities. This is done by grouping m neighboring values into a single bucket and setting each item’s probability to be the average of the m original probabilities. This bucketing process effectively reduces the CDF table size by a factor of m . We visualize the PMFs of discrete staircase Gaussian and discrete Gaussian in Fig. 7a. A formal definition of the discrete staircase DP noise is in Definition 4.

The staircase hyper-parameter m controls the utility-efficiency trade-off. As a larger m further reduces the size of the CDF table and improves the sampling efficiency, but the distribution deviates more from commonly-used DP noise with analytical form, which would complicate the worst-case analysis. We discuss selecting the staircase hyper-parameters in concrete applications in § 6.2.1.

Definition 4 (Staircase Discrete DP Noise Distribution). *Consider a symmetric discrete noise distribution $\mathcal{D}$ with PMF $p(x), x \in \mathbb{Z}$, the staircase noise distribution $\mathcal{D}_s$ ’s PMF $p_s(x)$ is defined as follows:*

$$p_s(x) = \frac{w_k}{m}, \quad k = \lfloor \frac{|x|}{m} \rfloor \cdot \text{SIGN}(x), \quad w_k = \sum_{r=0}^{m-1} p(|k|m + r)$$

where $m \in \mathbb{Z}_{>0}$ is the bucket width, w_k is the sum of the probability mass in the k -th bucket.

We now analyze discrete staircase DP noise’s privacy guarantees using Concentrated DP (CDP; see § 2). Theorem 2 formalizes the privacy guarantees of the staircase discrete noise. We show that by choosing the bucket size m such that the sensitivity Δ is multiple of m , the privacy loss of the staircase noise is upper-bounded by the original discrete noise. Intuitively, since staircase optimization “coarsens” the original discrete distribution by grouping probability mass into buckets, such coarsening reduces information leakage and provides stronger privacy guarantees [GV15]. We present the proof in Appendix C and discuss the generalization to multivariate scenarios in Appendix D.

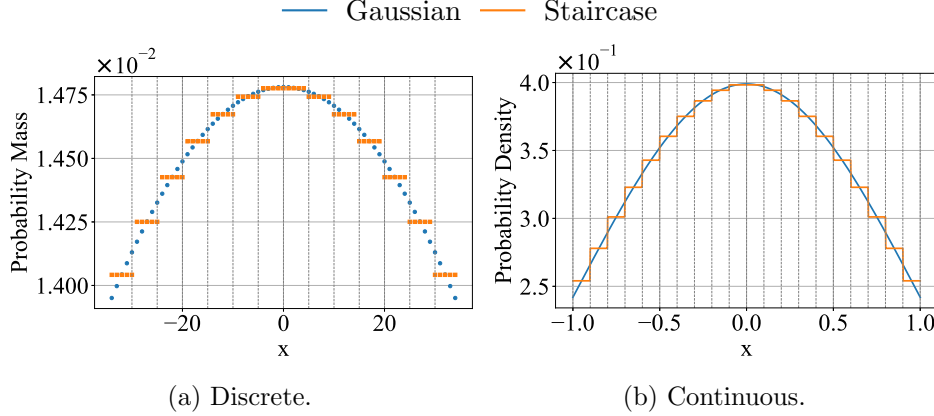


Figure 7: Discrete and continuous Gaussian and their staircase versions. For discrete distributions, $\sigma = 100$ and bucket width $m = 5$. For continuous distributions, $\sigma = 1$ and $m = 0.1$.

Theorem 2 (DP Guarantee of Staircase Discrete Noise). *If the original discrete DP noise $\mathcal{D}$ satisfies $\frac{1}{2}\epsilon^2$ -CDP, then the discrete staircase noise $\mathcal{D}_s$ also satisfies $\frac{1}{2}\epsilon^2$ -CDP, if the bucket width m satisfies $\Delta = tm$, $t \in \mathbb{Z}$, Δ is the sensitivity, and the original DP discrete noise distribution $\mathcal{D}$'s PMF is symmetric and monotonous, i.e., $p(x) = p(-x)$ and $p(x) \geq p(x+1)$, $\forall x \in \mathbb{Z}_{\geq 0}$.*

5.2 Noisette-SC: Discrete Staircase Sampling

The staircase DP noise can be easily sampled using CDF table lookup technique in NOISETTE-BASE. After transforming the original DP noise into a corresponding staircase version, the original truncation parameter M stay the same, while the lookup table size is reduced as it only needs to contain the CDF of the values that are multiple of m . The staircase DP noise thus reduces the table size by $m\times$, from M to $\frac{M}{m}$. To sample from the staircase distribution, we first sample an integer $x = km$, $k \in \mathbb{Z}_{\geq 0}$ from the CDF table. Note that x represents a sampled bucket index. Then, we uniformly sample another integer $y \in [0, m-1]$ and compute sample $z = x + y$, which follows the staircase discrete noise distribution. We present the detailed NOISETTE-SC protocol in Fig. 8, and similar to Fig. 6, the sign-bit sampling used for symmetric noise is omitted.

5.3 Noisette-Cont: Continuous (Floating Point) Noise Sampling

So far, we have shown staircase optimizations can significantly reduce the CDF lookup table size for discrete DP noise sampling. Now, we generalize our technique to floating-point noise sampling and introduce our protocol NOISETTE-CONT.

Definition 5 (Staircase Continuous DP Noise Distribution). *Consider a symmetric continuous noise distribution $\mathcal{D}$ with PDF $p(x)$, $x \in \mathbb{R}$ satisfying the $\frac{1}{2}\epsilon^2$ -CDP, the staircase noise distribution $\mathcal{D}_s$'s PDF $p_s(x)$ is defined as follows:*

$$p_s(x) = \frac{w_k}{m}, \quad k = \lfloor \frac{|x|}{m} \rfloor \cdot \text{SIGN}(x), \quad w_k = \int_{|k|m}^{(|k|+1)m} p(x) dx$$

where $m \in \mathbb{R}_{>0}$ is the bucket width, w_k is the sum of the probability density in the k -th bucket.

We visualize the staircase continuous Gaussian noise in Fig. 7b. Similar to the discrete Gaussian, we assign the values in each bucket with the same probability, which is the average of their

Protocol $\Pi_{\text{sampleStaircase}}$

Input:

- Truncated noise distribution $\mathcal{D}'$, sample number N .
- Invoke system type, NOISETTE-SC for discrete and NOISETTE-CONT for continuous sampling.

Publicly Compute:

- $\mathcal{P}$ and $\mathcal{V}$ convert the noise distribution to its staircase approximation, and obtain its bucket size m and staircase bucket index distribution $\mathcal{D}''$ over bucket bases $\{0, m, 2m, \dots\}$. $m \in \mathbb{Z}$ if NOISETTE-SC is invoked and $m \in \mathbb{R}$ if NOISETTE-CONT is invoked.
- Determine the required bits n' to represent random values from $[0, m)$.

Protocol:

1. $\mathcal{P}$ and $\mathcal{V}$ send $(\text{sample}, \mathcal{D}'', N)$ to $\mathcal{F}_{\text{sample}}$. $\mathcal{P}$ obtains $\mathbf{x}'_i, \llbracket \mathbf{x}'_i \rrbracket$, and $\mathcal{V}$ obtains $\llbracket \mathbf{x}'_i \rrbracket$, $i \in [N]$. And $\mathbf{x}'_i \in \{0, m, 2m, \dots\}$.
2. $\mathcal{P}$ and $\mathcal{V}$ send $(\text{rand}, n'N)$ to $\mathcal{F}_{\text{rand}}$. $\mathcal{P}$ obtains U and $\llbracket U \rrbracket$ and $\mathcal{V}$ obtains $\llbracket U \rrbracket$, where $\llbracket U \rrbracket = \{\llbracket U_0 \rrbracket, \llbracket U_1 \rrbracket, \dots, \llbracket U_{N-1} \rrbracket\}$, $U_i \in \{0, 1\}^{n'}$ are interpreted as uniformly random values from $[0, m)$.
3. For all $i \in [N]$, $\mathcal{P}$ and $\mathcal{V}$ use $\mathcal{F}_{\text{ZK}}$ to prove $\mathbf{x}_i = \mathbf{x}'_i + U_i$.

Output: $\mathcal{P}$ receives $\mathbf{x}_i, \llbracket \mathbf{x}_i \rrbracket$ and $\mathcal{V}$ receives $\llbracket \mathbf{x}_i \rrbracket$, $i \in [N]$.

Figure 8: NOISETTE-SC and NOISETTE-CONT protocols for verified sampling discrete and continuous staircase DP noise.

probabilities summation. We can effectively discretize the continuous distributions and preserve their tail behaviors. For one-dimensional case or multi-dimensional case with independent sensitivities, similar to Theorem 2, we set the bucket size m such that the sensitivity $\Delta = km$, where $k \in \mathbb{Z}$. The privacy loss of the staircase noise is upper bounded by the original continuous DP noise.

For multivariate noise of dimension d , it is non-trivial to directly generalize Theorem 4 in Appendix D to bound the privacy loss, because the sensitivity assigned to each dimension can be infinitely small for continuous cases. Computing the actual privacy cost is also challenging due to the large space of continuous value representations such as FP32. To address this challenge, we instead bound the privacy loss in CDP and present the following theorem. The high-level intuition is that the divergence is monotonous to the sensitivity $\Delta_i, i \in [d]$ of each dimension i (see our proof in Appendix C). Thus, if we add a little bit more sensitivity to each dimension to ensure that $\Delta_i = km, k \in \mathbb{Z}$, then Theorem 2 can be generalized, and we can upper bound the privacy loss with the original DP mechanism. A small increase in sensitivity leads to a slight compromise in privacy guarantees, but we can maintain the original privacy loss by choosing slightly larger noise scales.

Theorem 3 (DP Guarantee of Multivariate Continuous Staircase Noise). *The continuous staircase noise $\mathcal{D}_s$ satisfies $\frac{1}{2}(1+c)^p \epsilon^2$ -CDP, if the bucket width m satisfies $m = c\Delta_p/d^{1/p}$, where c is a*

hyper-parameter and Δ_p denotes the L_p norm sensitivity bound, and the original DP continuous noise distribution $\mathcal{D}$'s PDF is symmetric and monotonous, i.e., $p(x) = p(-x)$ and $p(x) \geq p(y)$, $\forall y \geq x \geq 0$.

With Theorem 3 (see proof in Appendix E), the total privacy loss is $(1 + c)^{p/2}\epsilon$. The CDF table size is linear to $1/c$. Thus, c can be set to balance the privacy guarantees and sampling efficiency. To maintain the same ϵ as the original DP mechanism, we take $\epsilon' = \frac{\epsilon}{(1+c)^{p/2}}$, and then calculate the original DP noise scale that satisfies (ϵ', δ) -DP, and then construct the staircase distribution and the CDF table accordingly.

The continuous noise sampling protocol NOISSETTE-CONT is similar to the discrete noise sampling protocol NOISSETTE-SC. We first sample $x = km, k \in \mathbb{Z}_{\geq 0}$ from the staircase CDF table, where k represents the sampled bucket index. Then we sample a continuous value $y \in [0, m)$. In the concrete implementation, real values are not representable for finite computers, and y is represented using floating-point. We first sample uniformly random bits using Π_{rand} as the mantissa and set constant exponent such that the value is uniform in $[0, 1)$. Then we compute the product of the sampled uniform and the public m to obtain y .

For symmetric noise optimization, we sample a sign indicator $s \in \{-1, 1\}$, and compute the continuous staircase noise as $z = s \cdot (x + y)$. Similar to Fig. 6, we omit this step for clarity in Fig. 8.

Discussion on continuous value representation. In practice, computers cannot represent truly continuous values; instead, they rely on finite-precision formats such as the FP32 representation defined by the IEEE 754 standard [Kah96]. This limitation has been shown to introduce subtle privacy leakage channels in floating-point arithmetic, and prior work has investigated both risks and possible mitigation strategies [Mir12].

It is important to note that addressing floating-point side channels is not the primary goal of this paper. Our system, NOISSETTE-CONT, samples values in a manner consistent with how they are realized on a finite-precision machine. In other words, NOISSETTE-CONT reflects the standard computational model without introducing additional sources of leakage beyond what arises inherently from floating-point arithmetic. For applications where floating point leakage is a concern, users can adopt existing countermeasures (e.g. the techniques proposed by [Mir12]) alongside NOISSETTE-CONT.

Security Analysis. The protocols NOISSETTE-SC and NOISSETTE-CONT are obtained by invoking NOISSETTE-BASE on a reduced public CDF table T to certify sampling of the (bucket) value $\mathbf{x}'_i$, and then applying a (proved-in-ZK) post-processing step that draws an independent uniform offset $u_i \in [0, m)$ (and a sign bit) via $\mathcal{F}_{\text{rand}}$ and outputs $\mathbf{x}_i = \mathbf{x}'_i + u_i$ (and $s_i \cdot \mathbf{x}_i$). Hence, in the $(\mathcal{F}_{\text{ZK}}, \mathcal{F}_{\text{rand}})$ -hybrid model with malicious prover and verifier, completeness, soundness, and zero-knowledge follow exactly as in § 4. In terms of DP guarantees, the only substantive change is replacing $\mathcal{D}$ by its staircase approximation, whose guarantees are captured by Theorem 2 and Theorem 3; in particular, NOISSETTE-SC preserves the original $\frac{1}{2}\epsilon^2$ -CDP, while NOISSETTE-CONT incurs at most the stated multiplicative CDP factor, which we compensate by instantiating the base noise with $\epsilon' = \epsilon/(1 + c)^{p/2}$. Consequently, all residual truncation, precision, and certification errors are absorbed as an additive increase to the final δ parameter exactly as in § 4.

6 Evaluation

In this section, we evaluate the performance of our protocols. In § 6.1, we benchmark the performance of NOISSETTE-BASE and evaluate its scalability w.r.t. number of samples and table size. We then compare NOISSETTE-BASE with state-of-the-art (SOTA) protocols on discrete Gaussian and Laplace

sampling. In § 6.2, we show that our staircase optimizations used in NOISSETTE-SC and NOISSETTE-CONT can further improve efficiency and enable efficient continuous DP noise sampling with the same privacy guarantees and minor influences on utility in concrete applications. We consider two widely deployed DP applications: mean estimation [DMNS06b] and federated learning (FL) [AKL21] and also compare with SOTA baselines.

Evaluation setup. We implement NOISSETTE using EMP-toolkit [WMK16] for the interactive zero-knowledge interface. The experiments for benchmarking NOISSETTE-BASE are conducted on AWS instance type `c5.xlarge`, and the experiments for evaluating the utility and efficiency of NOISSETTE-SC and NOISSETTE-CONT across concrete DP applications are conducted on AWS instance type `p3.16xlarge`. We consider two network settings:

- **LAN:** Bandwidth up to 1 Gbps and latency 0.1 ms.
- **WAN:** Bandwidth up to 100 Mbps and latency 100 ms.

Baselines. As mentioned in § 1, most prior work targets a specific noise distribution. Accordingly, for discrete distributions, we evaluate NOISSETTE-BASE against SOTA oblivious sampling protocols for both discrete Gaussian and Laplace. For each distribution, we consider two classes of baselines: (1) the SOTA sampling protocols for discrete Gaussian and Laplace proposed in [WYF⁺23], denoted as PRIOR-DISC; and (2) the Bernoulli sampling protocol from [CsU19], instantiated with an oblivious stack (Ostack) and combined with PRIOR-DISC as demonstrated in [FW24], denoted as PRIOR-DISC-OSTACK.

We note that these baseline methods were originally designed as oblivious sampling circuits for discrete Laplace and Gaussian distributions in the setting of secure multi-party computation, rather than ZKP. Nevertheless, their techniques can be readily adapted to the ZKP setting, and they represent the most efficient known sampling circuits, thereby establishing strong SOTA baselines.

For continuous cases, we compare NOISSETTE-CONT against the SOTA approach for certifiable floating-point Gaussian sampling [STC⁺24], denoted as PRIOR-CONT.

6.1 Discrete DP Noise Sampling Benchmark

We first benchmark the performance of NOISSETTE-BASE. Then we compare it with SOTA baselines PRIOR-DISC and PRIOR-DISC-OSTACK on sampling discrete Gaussian and Laplace noise.

6.1.1 Scalability of Noisette-Base

We demonstrate the scalability of NOISSETTE-BASE by evaluating its runtime and communication cost under various numbers of samples and table sizes.

Scalability w.r.t. number of samples. We benchmark the runtime and communication overhead of NOISSETTE-BASE for a discrete distribution with CDF table size $M = 10^4$ where each probability uses $n = 64$ bits—sufficient for most practical DP applications. As shown in Fig. 9a, the per-sample runtime and communication costs decrease as the sample number N increases, reflecting the amortization analyzed in § 4.

Scalability w.r.t. table size. Fig. 9b illustrates the cost of sampling $N = 10^4$ values from discrete distributions with varying CDF table sizes M . The cost scales linearly with M , driven by the linear increase in $\mathbb{F}_{ZK}$ multiplications.

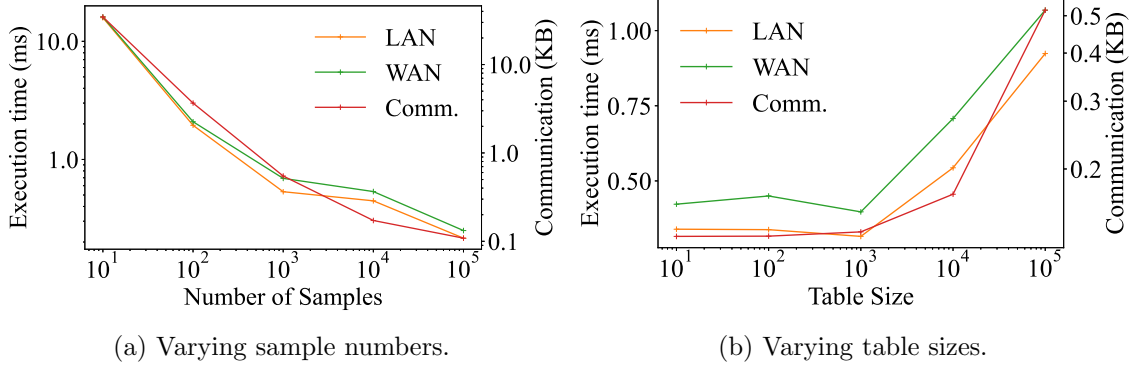


Figure 9: Benchmarks of NOISSETTE-BASE. The runtime (ms) and communication (KB) required per sample when (a) sampling from a discrete distribution with CDF table size $M = 10^4$, (b) drawing $N = 10^4$ samples with varying CDF table sizes.

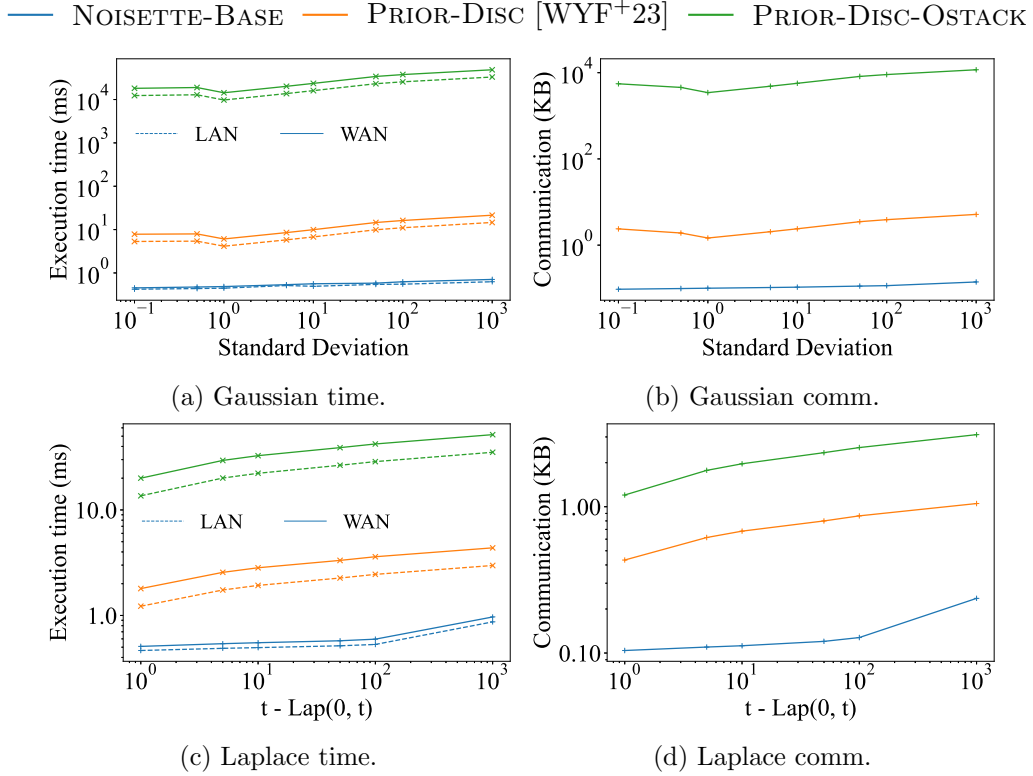


Figure 10: Comparison with SOTA baselines. Table size is determined by noise parameters. Number of samples $N = 2^{14}$.

6.1.2 Comparison With Prior Work

We compare NOISSETTE-BASE with SOTA sampling protocols for discrete Gaussian and Laplace, specifically PRIOR-DISC [CsU19] and PRIOR-DISC-OSTACK [FW24]. Using the prior work’s MPC implementation¹, we obtain the circuit AND gate counts to estimate the costs. Fig. 10 details the

¹<https://github.com/yuchengxj/Secure-sampling-benchmark>

comparison across both network settings with sample size $N = 2^{14}$.

Discrete Gaussian. NOISSETTE-BASE outperforms both rejection sampling protocols in all settings (Fig. 10a and 10b), demonstrating a $10 - 30\times$ runtime improvement and $10 - 36\times$ communication reduction compared to PRIOR-DISC [WYF⁺23]. Specifically, for $\sigma = 1000$ in the LAN setting, our protocol requires $23\times$ less time and $36\times$ less communication than PRIOR-DISC. Furthermore, we achieve a $52\times$ (resp. $83\times$) improvement in runtime (resp. communication) compared to discrete Gaussian sampling with the Ostack algorithm.

Discrete Laplace. NOISSETTE-BASE also significantly outperforms PRIOR-DISC [WYF⁺23] for discrete Laplace distributions. For $t = 100$, NOISSETTE-BASE yields a $6\times$ improvement in both time and communication. Compared to PRIOR-DISC-OSTACK, we observe a $17\times$ improvement in runtime and $19\times$ in communication. Note that the performance gap diminishes as the noise scale increases; the cost of prior work is proportional to $\log(M)$, whereas our protocol scales linearly with M . This motivates our staircase optimizations for large CDF tables.

As expected, for both discrete Gaussian and Laplace, NOISSETTE-BASE greatly outperforms prior work by avoiding large circuits for complex noise sampling, while set inclusion is executed more efficiently and further benefits from a large number of samples.

6.2 Evaluation on Concrete Applications

We evaluate the utility and efficiency of our discrete and continuous DP noise sampling protocols on two widely deployed DP applications: privacy-preserving mean estimation and differentially private federated learning (DPFL). We measure runtime and communication costs alongside utility metrics to assess the utility-efficiency tradeoff. Detailed settings for each use case are provided below.

6.2.1 Privacy Preserving Mean Estimation

Privacy-preserving mean estimation is a fundamental application of DP, widely used in real-world scenarios and as a benchmark for DP protocols [FW24, AKL21]. In this section, we evaluate the utility and efficiency of NOISSETTE-BASE, NOISSETTE-SC, and NOISSETTE-CONT for mean estimation. We assess scalability by measuring performance w.r.t. the number of noise samples and the privacy budget (table size). Furthermore, we analyze the impact of staircase hyper-parameters in NOISSETTE-SC and NOISSETTE-CONT to demonstrate the utility-efficiency tradeoff of our staircase optimizations.

Applications setup. To simulate a real-world scenario, we consider s datasets $X_i \in \mathbb{Z}^{h \times d}, i \in [s]$. These datasets represent smart meter data collected from s neighborhoods. Each neighborhood contains h devices, where each data point is a d -dimensional vector. Smart meters commit these data samples and send them to an electricity company for aggregation. The aggregator computes the average of the collected data and applies DP mechanisms before releasing the aggregated electricity usage statistics. Crucially, the aggregator proves to a regulator that the DP mechanism is executed correctly, ensuring the privacy of the released statistics is rigorously protected. In total, the aggregator generates $s \cdot d$ DP noise samples.

We synthesize the s datasets $X_i \in \mathbb{R}^{h \times d}$ by randomly sampling $h = 10000$ data points from a multivariate Gaussian $\mathcal{N}(0, I_d)$ with $d = 1000$, normalizing each feature to $[0, 1]$. For discrete DP scenarios, we quantize data samples using fixed-point quantization with a scale factor of $2^\ell, \ell = 4$ for all dimensions.

To evaluate utility, we measure the mean squared error (MSE) between the certified DP mean estimation and the ground truth mean. To evaluate efficiency, we measure the runtime and communication costs of NOISSETTE in the WAN setting. Without loss of generality, we primarily

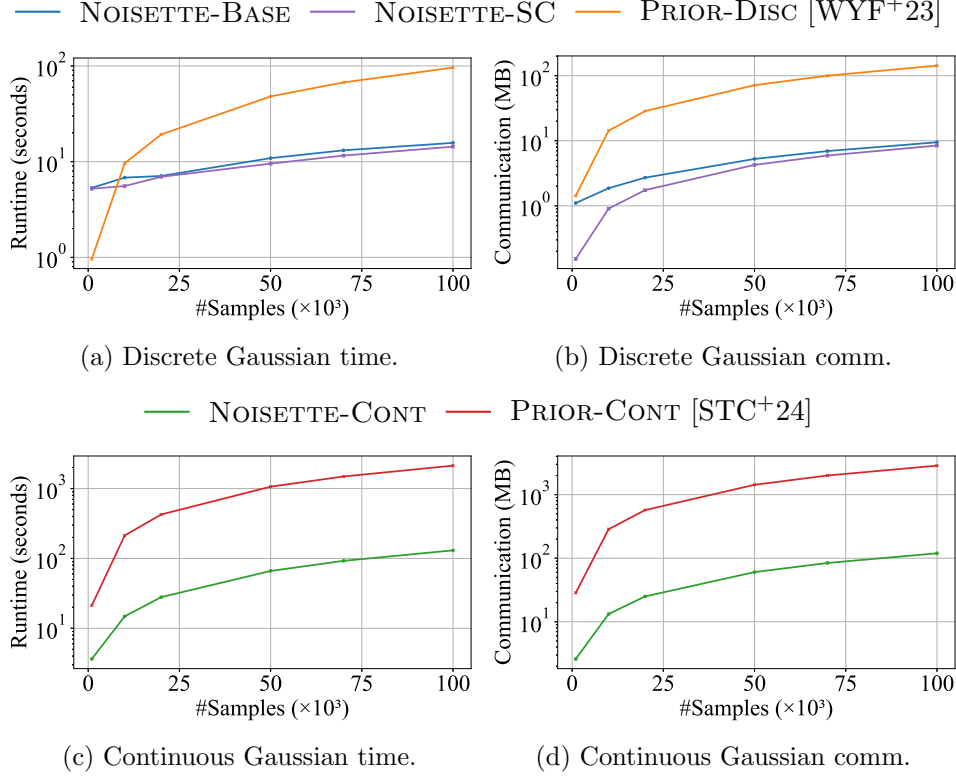


Figure 11: Runtime and communication v.s. sample numbers for discrete and continuous Gaussian in mean estimation.

evaluate *continuous* and *discrete* Gaussian mechanisms given their widespread adoption; however, these findings generalize to any additive noise DP mechanism.

Scalability w.r.t. noise sample numbers. The total number of noise samples, $s \times d$, is a key factor influencing sampling efficiency. We evaluate this by measuring runtime and communication costs across different s . We generate discrete and continuous Gaussian noise to ensure an (ϵ, δ) -DP guarantee, using $\epsilon = 0.5, \delta = 10^{-6}$. We compare the costs against SOTA baselines PRIOR-DISC [CsU19] and PRIOR-CONT [STC+24], with results presented in Fig. 11.

In Fig. 11a and Fig. 11b, we compare the runtime and communication costs of our protocols against the prior SOTA baseline for discrete Gaussian sampling [CsU19]. We set the staircase bucket size to $m = 16$ (selection of hyper-parameters is discussed in the following section). Our results align with the findings in § 6.1. With a larger number of samples, our protocols are significantly more efficient than prior work, achieving approximately $7\times$ faster runtime and $17\times$ lower communication. We observe that for very small sample sizes (1000), our protocols are slightly slower; however, our technique efficiently batches large numbers of samples, demonstrating superior scalability. We also observe that staircase optimization further improves performance, particularly when the sample number is small. As expected, with increasing sampling numbers, the savings of staircase optimizations are less significant, as the sample number dominates the costs.

We observe similar trends for continuous Gaussian sampling, as shown in Fig. 11c and Fig. 11d. Using a staircase hyperparameter of $c = 0.05$, NOISETTE-CONT is consistently more efficient than PRIOR-CONT [STC+24], with approximately $16\times$ faster runtime and $24\times$ less communication.

Scalability w.r.t. privacy budget ϵ . Privacy budget ϵ is another factor that controls the utility,

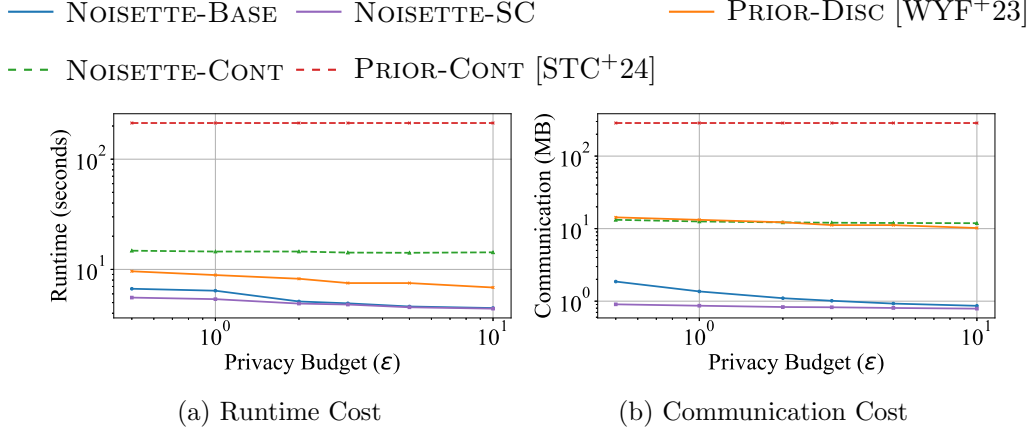


Figure 12: Runtime and communication cost v.s. ϵ for discrete and continuous Gaussian in mean estimation.

Mechanism	ϵ	δ	MSE
Discrete Gaussian	0.5	10^{-6}	7.96×10^{-4}
	1		1.88×10^{-4}
	10		3.26×10^{-6}
Discrete Gaussian with NOISSETTE-BASE	0.5	10^{-6}	7.95×10^{-4}
	1		2.01×10^{-4}
	10		3.17×10^{-6}
Discrete Gaussian with NOISSETTE-SC	0.5	10^{-6}	7.44×10^{-4}
	1		2.22×10^{-4}
	10		3.23×10^{-6}
Continuous Gaussian	0.5	10^{-6}	7.51×10^{-4}
	1		2.06×10^{-4}
	10		3.38×10^{-6}
Continuous Gaussian with NOISSETTE-CONT	0.5	10^{-6}	7.94×10^{-4}
	1		2.13×10^{-4}
	10		3.73×10^{-6}

Table 2: Utility (MSE) of mean estimation under various ϵ .

privacy, and efficiency tradeoff. A smaller ϵ implies stronger privacy protection, requiring a larger noise scale and CDF table, which increases sampling costs. In this experiment, we analyze the sampling efficiency and utility of our protocols under varying ϵ , with 10000 noise samples.

Fig. 12 presents the runtime and communication costs. As expected, increasing ϵ reduces the CDF table size, thereby lowering the cost of our sampling protocols. While the sampling cost of prior work remains relatively consistent regardless of noise scale, our protocols demonstrate more significant savings as ϵ increases as the CDF table size becomes smaller.

We further evaluate the utility of our protocols in Tab. 2. We achieve utility comparable to DP noise without certifications, demonstrating the high utility of our protocols.

Utility-Efficiency tradeoff on staircase hyper-parameters. As discussed in § 5, the hyper-parameter controlling bucket size (m in NOISSETTE-SC and c in NOISSETTE-CONT) balances the utility and efficiency tradeoff. We set $s = 1$, $\epsilon = 0.1$, and $\delta = 10^{-6}$, measuring the cost of our

Mechanism	Staircase Parameter	MSE
Discrete Gaussian	–	1.65×10^{-2}
Discrete Gaussian with NOISSETTE-SC	$m = 2$	1.54×10^{-2}
	$m = 4$	1.58×10^{-2}
	$m = 8$	1.65×10^{-2}
	$m = 16$	1.62×10^{-2}
Continuous Gaussian	–	1.72×10^{-2}
Continuous Gaussian with NOISSETTE-CONT	$c = 0.01$	1.74×10^{-2}
	$c = 0.05$	1.78×10^{-2}
	$c = 0.1$	2.05×10^{-2}
	$c = 0.3$	2.69×10^{-2}

Table 3: Utility (MSE) of mean estimation using of NOISSETTE-SC and NOISSETTE-CONT under varying staircase parameters with fixed $\epsilon = 0.1$, $\delta = 10^{-6}$, $s = 1$ and $d = 10^3$.

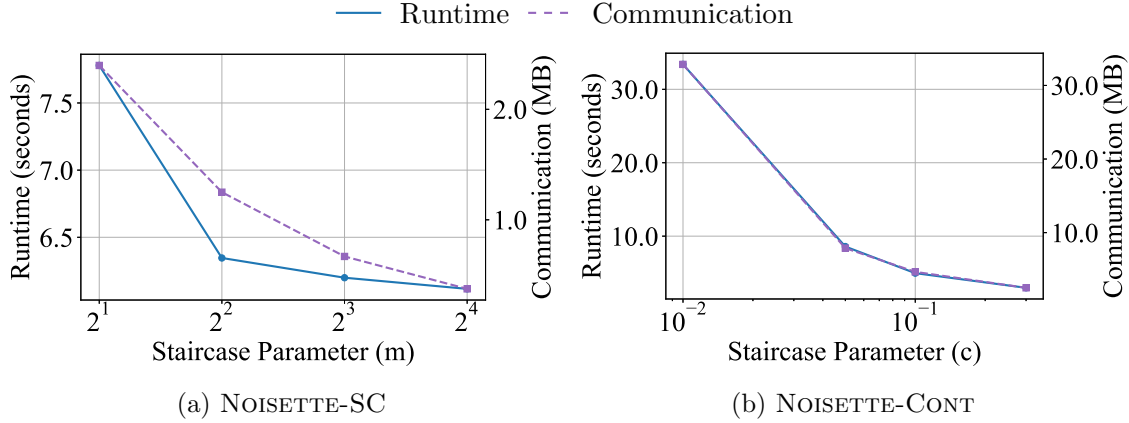


Figure 13: Runtime and communication cost of NOISSETTE-SC and NOISSETTE-CONT v.s. the respective staircase hyper-parameters with fixed $\epsilon = 0.1$, $\delta = 10^{-6}$, $d = 10^3$ and $s = 1$.

protocols and the MSE of the DP aggregations compared to the ground truth under different staircase parameters.

We present the cost in Fig. 13. As expected, increasing the staircase parameters (m for NOISSETTE-SC and c for NOISSETTE-CONT) significantly reduces runtime and communication costs. Larger parameters reduce the size of the CDF table, making lookups more efficient. We further evaluate the utility tradeoff by measuring the MSE with different staircase parameters in Tab. 3. Increasing m for NOISSETTE-SC has minimal influence on utility, yielding results comparable to discrete Gaussian without staircase optimizations. For NOISSETTE-CONT, the staircase modifies the distribution more as parameter c increases. A large c increases MSE and reduces utility; in this experiment, $c = 0.05$ achieves a great balance between utility and efficiency. In practice, these parameters can be tuned to specific applications to achieve the desired balance.

6.2.2 Differentially Private Federated Learning

DPFL has emerged as a crucial real-world application of DP, with widespread adoption by industry leaders such as Apple and Google for protecting user data privacy [HRM⁺18, PSM⁺21]. We demonstrate the scalability and generalizability of NOISSETTE by training large DPFL models.

Application setup. We simulate a FL environment with a central server and 100 clients. Each client holds a private dataset. The server and clients collaboratively train a neural network with 223,614 parameters using DPSGD [AKL21, XZA⁺23]. In each training round, we randomly select 10 clients to participate. We evaluate model accuracy on the EMNIST dataset [CATVS17], a standard handwritten digits and characters image classification benchmarking dataset with 62 classes. We randomly split the training set of 731,668 samples into 100 clients and evaluate the model accuracy on the test set of 82,587 samples. We consider two widely studied privacy settings:

1. **Central DP:** Clients trust the server with plaintext gradients. The server aggregates these and adds DP noise before the model update.
2. **Local & Central DP:** Clients do not fully trust the server. They add local DP noise to their gradients before transmission; these are aggregated at the server, which then adds a second layer of noise to strengthen the central DP.

These two settings correspond to different levels of trust in the server. In the Central DP setting, clients place greater trust in the server, which has access to their plaintext gradients. In contrast, in the Local & Central DP setting, clients do not wish to directly disclose their gradients to the server and thus apply an additional layer of local DP. And locally added noise is aggregated on the server to provide a central DP guarantee.

In both scenarios, parties that add DP noise use NOISETTE to generate proofs of correct noise sampling for a verifier (e.g., a regulator). For the Central DP setting, we use the *continuous* (floating-point) Gaussian mechanism. For the Local & Central DP setting, we adopt a *discrete* approach where gradients are quantized and noise is added via the Skellam mechanism [AKL21]. The Skellam mechanism is particularly suited for FL as it is closed under summation, providing tighter privacy guarantees than the discrete Gaussian. Notably, NOISETTE is the first to support certifiable Skellam sampling, highlighting its compatibility with arbitrary additive noise mechanisms.

For Central DP, the server adds Gaussian noise to achieve (ϵ, δ) -DP with $\epsilon = 10, \delta = 10^{-6}$. For Local & Central DP, clients and server collectively achieve the same (ϵ, δ) central DP via Skellam noise. To reflect real-world deployments, we assign asymmetric resources: the server uses an AWS `c6i.32xlarge` (64 threads), while clients use 4 threads each. We train for 100 rounds in a WAN environment.

Central DP results. We utilize NOISETTE-CONT to sample floating-point Gaussian noise. We evaluate the influence of the staircase hyper-parameter c that controls the utility-efficiency tradeoff and compare with PRIOR-CONT [STC⁺24].

Tab. 4 shows that NOISETTE-CONT achieves a $64\times$ speedup and $15\times$ lower communication overhead compared to the prior SOTA. Interestingly, varying c has minimal impact on efficiency in this large-scale setting. Because we draw $N = 223,614$ samples per iteration, as discussed in § 4 the total cost $O(M + c'N)$ is dominated by N rather than the table size M . Fig. 14 confirms that for $c \leq 0.5$, NOISETTE-CONT maintains accuracy comparable to uncertified Gaussian mechanisms, demonstrating its high utility.

Local & Central DP results. In this setting, we evaluate our discrete noise sampling protocol NOISETTE-BASE on Skellam mechanism. Since the gradients are quantized, the quantization bit length ℓ impacts the model accuracy and affects the CDF table size. We evaluate the model accuracy w.r.t. the central privacy budget ϵ spent with varying ℓ as shown in Fig. 15. The model accuracy increases with ℓ as the values are represented with higher precision.

With more privacy budget ϵ spent, the model accuracy increases for all values of ℓ . Our model accuracy matches the uncertified Skellam mechanism in [AKL21], where they train a $4\times$ larger model with 1500 rounds and achieve more than 60% accuracy, and our accuracy exceeds 50% in 100

DP Noise	Sampling Time (s)	Sampling Communication (MB)
PRIOR-CONT [STC ⁺ 24]	3637.04	6145.18
NOISETTE-CONT $c = 0.2$	59.33	422.31
NOISETTE-CONT $c = 0.5$	57.14	407.93
NOISETTE-CONT $c = 0.8$	57.19	406.18
NOISETTE-CONT $c = 1.0$	56.77	405.71

Table 4: Noise sampling cost of SOTA baseline and NOISETTE-CONT in DPFL with varying c in Central DP setting.

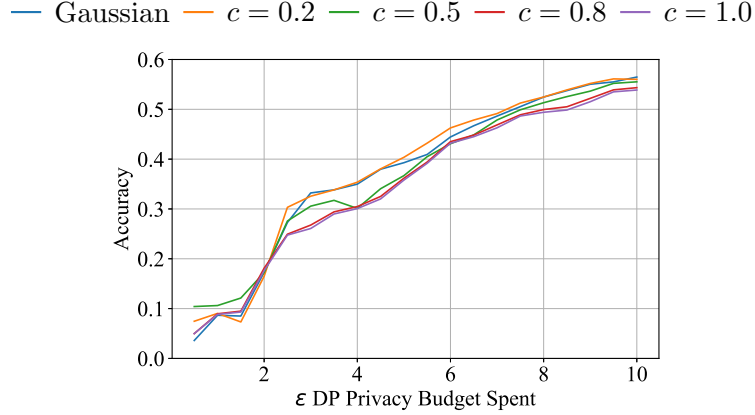


Figure 14: Model accuracy w.r.t. the DP privacy budget ϵ spent for continuous Gaussian and staircase Gaussian with different hyper-parameters c in the Central DP Setting.

Cost	Quantization Bit ℓ	Client	Server	Total
Runtime	11	31.11 s	9.80 s	40.91 s
	12	32.12 s	9.92 s	42.04 s
	13	34.25 s	12.29 s	46.54 s
	14	37.28 s	12.53 s	49.81 s
Communication	11	19.20 MB	25.87 MB	45.07 MB
	12	19.81 MB	27.55 MB	47.36 MB
	13	20.47 MB	32.98 MB	53.45 MB
	14	21.30 MB	52.81 MB	74.11 MB

Table 5: Runtime and communication cost for DPFL with different quantization bits ℓ in the Local & Central DP setting. *Since NOISETTE is the first protocol to support verifiable Skellam mechanism, we do not compare with prior work here.*

rounds. This underscores NOISETTE’s ability to maintain high utility in large-scale applications while providing the first certifiable implementation for Skellam noise, demonstrating NOISETTE can be generalized to *any* additive DP noise.

Sampling costs are reported in Tab. 5. While the CDF table size scales with 2^ℓ , the cost increases only moderately because the total overhead is dominated by the large number of noise samples (model parameters) drawn each round. Even for the largest table size ($\ell = 14$), the sampling cost remains efficient, confirming NOISETTE is well-suited for high-dimensional use cases.

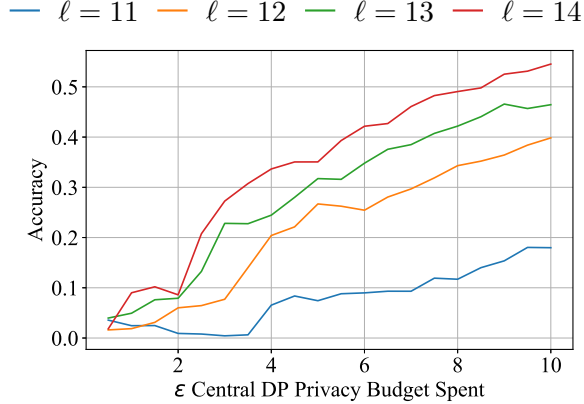


Figure 15: Model accuracy w.r.t. central DP ϵ spent for various quantization bit-length ℓ in the Local & Central DP setting.

7 Related Work

Our work intersects with several areas of research in cryptography and privacy-preserving machine learning. We organize the related work into two main categories: certifying differential privacy and differential privacy in MPC.

Certifying Differential Privacy. Prior work has explored the intersection of differential privacy and cryptographically verifiable computation. [STC⁺24] proposed a framework for certifying the training of DPSGD using the Box-Muller transform [BM58] to sample floating-point Gaussian noise, which involves costly computations like square root and log functions. [NFPH15] also proposed a framework for certifying Laplace noise sampling, which involves evaluating complex log functions in ZK.

Recent work [BGKW24, BAT24] proposed protocols for certifying simple discrete noise distributions like Binomial, but left the certification of more complex distributions like discrete Laplace and Gaussian as an open challenge. NOISETTE addresses this gap by providing a generic protocol for certifying arbitrary discrete noise sampling.

Another line of work [NST⁺21, NHS⁺23] focuses on analyzing the tightness and correctness of the DP algorithms, especially for DP-SGD. Their methods can quantify the robustness of the DP algorithms and the correctness of the implementations, however, their results are not cryptographically provable. Their verification usually considers a strong adversary that can perform membership inference attacks and imply the access to the sensitive datasets, making the auditing methods inapplicable to privacy-sensitive scenarios.

Differential Privacy in MPC. Local DP requires each client to add some noise while central DP requires the clients to trust the server for adding the noise. To get the best of both worlds, several works propose the use of MPC for sampling noise from specific distributions. Dwork et al. [DKM⁺06] describes an efficient method for sampling from a geometric distribution using securely sampled biased coins. Champion et. al. [CsU19] proposes a method for securely sampled biased coins based on an efficient oblivious stack protocol. Wei et al. [WYF⁺23] designs a rejection sampling protocol for discrete Gaussian distribution based on the method from [CKS20], using the Geometric sampling from [DKM⁺06]. Fu and Wang [FW24] show a comprehensive study of differential privacy techniques in MPC. They also evaluate the rejection sampling protocol from [WYF⁺23] with the oblivious stack based biased coin sampling [CsU19].

[KMS⁺24] proposes algorithms for sampling from discrete Laplace and Gaussian distributions, snapping mechanism used to handle the privacy issues that arise due to floating point operations and integer scaling mechanisms for both the distributions. [FFR⁺25] proposes a generic MPC protocol for any DP mechanism using MPC using lookup tables where the distribution is truncated and represented as a table and a random entry from the table is sampled with equal probability. However, they use threshold homomorphic encryption to realize their algorithm efficiently in MPC, thus, it is non-trivial to adapt their approach to a ZK setting.

The approaches from prior work that use MPC interface to evaluate the circuit for efficient sampling can be easily adapted to ZK setting to achieve certifiable DP by using a ZK interface to prove the circuit satisfiability. However, these circuits are typically large and thus, do not scale well for a large number of samples. A concurrent work from [MRS25] also explores the use of LUT for noise sampling in MPC. However, their implementation differs from ours: they approximate the table index distribution using a Bernoulli distribution and perform the LUT evaluation within the MPC protocol. In contrast, NOISETTE employs inverse transform sampling, achieving concrete efficiency for large sample batches in ZKP. Furthermore, while adapting their method to floating-point distributions is non-trivial, NOISETTE supports them naturally.

8 Conclusion

We propose NOISETTE, a suite of simple yet powerful protocols for certifying differential privacy noise sampling that support any continuous or discrete DP distribution. In contrast to prior work, which focuses on optimizing specific mechanisms, NOISETTE provides a general and unifying framework that achieves order-of-magnitude improvements in both computational efficiency and communication overhead. We further validate its practicality by integrating NOISETTE into real-world DP applications, including mean estimation and federated learning for privacy-preserving model training. Our results establish NOISETTE as the first efficient, scalable, and general-purpose solution for certifiable DP noise sampling.

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A DP Mechanisms

In this section, we introduce the commonly used DP noise, including (discrete) Laplace, (discrete) Gaussian and Skellam [DR⁺14, CKS20, GRS08, AKL21, BW18].

- **Laplace Mechanism [DR⁺14]:** For a function $f : \mathcal{D} \rightarrow \mathbb{R}$ with sensitivity Δ , the mechanism is defined as $\mathcal{M}(x) = f(x) + \text{Lap}(t)$, where $\text{Lap}(t)$ denotes the Laplace distribution with mean 0 and scale parameter t . Its probability density function is:

$$\Pr[Y = y] = \frac{1}{2t} \exp\left(-\frac{|y|}{t}\right), \quad y \in \mathbb{R}.$$

The Laplace mechanism satisfies pure ε -DP, where $\varepsilon = \frac{\Delta}{t}$.

- **Gaussian Mechanism [BW18]:** For a function $f : \mathcal{D} \rightarrow \mathbb{R}$ with sensitivity Δ , continuous Gaussian mechanism is defined as $\mathcal{M}(x) = f(x) + \mathcal{N}(0, \sigma^2)$, where $\mathcal{N}(0, \sigma^2)$ is the Gaussian distribution with mean 0 and variance σ^2 . Its probability density function is:

$$\Pr[Y = y] = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2}{2\sigma^2}\right), \quad y \in \mathbb{R}.$$

The Gaussian mechanism satisfies (ε, δ) -DP, with

$$\delta = \Phi\left(\frac{\Delta}{2\sigma} - \frac{\epsilon\sigma}{\Delta}\right) - e^\epsilon \Phi\left(-\frac{\Delta}{2\sigma} - \frac{\epsilon\sigma}{\Delta}\right),$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. In practice, a commonly used classical bound is

$$\sigma = \frac{\Delta}{\epsilon} \sqrt{2 \log \frac{1.25}{\delta}}.$$

- **Discrete Laplace Mechanism [GRS08]:** For a function $f : \mathcal{D} \rightarrow \mathbb{Z}$ with sensitivity Δ , the mechanism is defined as $\mathcal{M}(x) = f(x) + \text{Lap}_{\mathbb{Z}}(t)$, where $\text{Lap}_{\mathbb{Z}}(t)$ denotes the discrete Laplace distribution. Its probability mass function is:

$$\Pr[Y = k] = \frac{e^{\frac{1}{t}} - 1}{e^{\frac{1}{t}} + 1} \cdot e^{-\frac{|k|}{t}}, \quad k \in \mathbb{Z}$$

Discrete Laplace mechanism satisfies pure ε -DP, where $\varepsilon = \frac{\Delta}{t}$.

- **Discrete Gaussian Mechanism [CKS20]:** For a function $f : \mathcal{D} \rightarrow \mathbb{Z}$ with sensitivity Δ , discrete Gaussian mechanism is defined as $\mathcal{M}(x) = f(x) + \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$, where $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ is the discrete Gaussian distribution with zero mean and variance σ^2 . Its probability mass function is:

$$\Pr[Y = k] = \frac{1}{\sum_{i \in \mathbb{Z}} e^{-\frac{i^2}{2\sigma^2}}} \cdot e^{-\frac{k^2}{2\sigma^2}}, \quad k \in \mathbb{Z}$$

Discrete Gaussian mechanism satisfies (ε, δ) -DP, with

$$\delta = \Pr_{Y \leftarrow \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)} \left[Y > \frac{\varepsilon \sigma^2}{\Delta} - \frac{\Delta}{2} \right] - e^\varepsilon \cdot \Pr \left[Y > \frac{\varepsilon \sigma^2}{\Delta} + \frac{\Delta}{2} \right].$$

- **Skellam Mechanism [AKL21]:** For a function $f : \mathcal{D} \rightarrow \mathbb{Z}$ with sensitivity Δ , the Skellam mechanism is defined as $\mathcal{M}(x) = f(x) + Sk(\mu)$, where $Sk(\mu)$ denotes the Skellam distribution, which is the difference of two independent Poisson variables with mean μ . Its probability mass function is:

$$\Pr[Y = k] = e^{-2\mu} \cdot \frac{\mu^{|k|}}{|k|!} \cdot I_{|k|}(2\mu), \quad k \in \mathbb{Z},$$

where I_k is the modified Bessel function of the first kind. Its privacy guarantee can be tightly bounded using Rényi-DP [Mir17], and it is particularly useful for distributed settings given that Skellam distribution is closed under summation [AKL21].

There are also other DP mechanisms with less tight privacy guarantees, such as the Binomial mechanism [DKM⁺06], which uses binomial noise to randomize the function output. Prior works have studied certifying the Binomial mechanism [BGKW24] and leave certifying more complex noise distributions like Laplace and Gaussian as challenging future work.

B Proof of Theorem 1

Proof. Fix $\mathcal{D}'$ and N , and let $T \leftarrow \Pi_{\text{TConstruct}}(\mathcal{D}', N)$ be the public table $T = \{(p_i^{\min}, p_i^{\max}, x_i)\}_{i \in [M]}$ such that the intervals $\{(p_i^{\min}, p_i^{\max}]\}_{i \in [M]}$ form a partition of $\{0, \dots, 2^n - 1\}$. Define the deterministic map $g_T : \{0, \dots, 2^n - 1\} \rightarrow \{x_1, \dots, x_M\}$ by $g_T(u) = x_j$ for the unique j satisfying $p_j^{\min} < u \leq p_j^{\max}$, and let $\widehat{\mathcal{D}'}$ be the induced distribution of $g_T(U)$ for uniform $U \in \{0, 1\}^n$ (interpreted as an integer in $\{0, \dots, 2^n - 1\}$).

We prove realization by considering three cases.

Case 1 (Both parties honest / completeness). By the specification of $\mathcal{F}_{\text{random}}$, the prover receives independent uniform coins $U_1, \dots, U_N \in \{0, 1\}^n$ and both parties obtain commitments $\llbracket U_i \rrbracket$. For each $i \in [N]$, the honest prover chooses the unique row $S_i := T_j$ such that $p_j^{\min} < U_i \leq p_j^{\max}$, and uses $\mathcal{F}_{\text{ZK}}$ to prove the relations in Step 5. Completeness of $\mathcal{F}_{\text{ZK}}$ implies $\mathcal{V}$ outputs **true**. Moreover, $\mathcal{P}$ outputs $S_{i,2} = g_T(U_i)$, hence $\{S_{i,2}\}_{i \in [N]}$ are i.i.d. from $\widehat{\mathcal{D}'}$, and $\mathcal{V}$ receives the corresponding commitments $\{\llbracket S_{i,2} \rrbracket\}_{i \in [N]}$. Finally, since T is obtained from n -bit fixed-point CDF probabilities for $\mathcal{D}'$, then by Lemma 2 the joint distribution satisfies $\Delta(\widehat{\mathcal{D}'}^N, (\mathcal{D}')^N) \leq \delta_2$. Equivalently, the outputs match N ideal samples from $\mathcal{D}'$ except with probability at most δ_2 .

Case 2 (Corrupted prover / soundness). Let $\mathcal{P}^*$ be any PPT prover and suppose $\mathcal{V}$ outputs **true**. By soundness of $\mathcal{F}_{\text{ZK}}$, except with probability $\text{negl}(\kappa)$, Step 5a implies $S_{i,0} < U_i \leq S_{i,1}$ for all $i \in [N]$. For Step 5b, if $\mathcal{P}^*$ causes acceptance while committing to some triple $(S_{i,0}, S_{i,1}, S_{i,2})$ that is not a row of T , then the compressed inclusion check requires a collision under the degree-2 map

Protocol Π_{lookup}

Input: $\mathcal{P}$ holds sets A and B . Both parties hold A and $\llbracket B \rrbracket$. $|A|$ is M and $|B|$ is N .

Protocol:

1. $\mathcal{P}$ computes $C = A \parallel B$ and sorts the elements in C according to A , i.e., for $i < i' \in [N+M]$ such that

- *Case 1.* $C[i] = C[i']$.
- *Case 2.* For $C[i] \neq C[i']$ where $C[i] = A[j]$ and $C[i'] = A[j']$ then $j < j'$.

$\mathcal{P}$ sends $\llbracket C \rrbracket$ to $\mathcal{V}$.

2. $\mathcal{P}$ defines two polynomials described in [GW20]:

$$\begin{aligned} f(\beta, \gamma) &:= (1 + \beta)^N \prod_{i \in [N]} (\gamma + B[i]) \\ &\quad \cdot \prod_{i \in [M-1]} (\gamma(1 + \beta) + A[i] + \beta A[i+1]) \\ g(\beta, \gamma) &:= \prod_{i \in [N+M-1]} (\gamma(1 + \beta) + C[i] + \beta C[i+1]) \end{aligned}$$

3. $\mathcal{V}$ samples $a, b \leftarrow \mathbb{F}_{2^k}$ and sends it to $\mathcal{P}$.
4. $\mathcal{P}$ and $\mathcal{V}$ compute $\llbracket f(a, b) \rrbracket$ and $\llbracket g(a, b) \rrbracket$.
5. $\mathcal{P}$ opens $\llbracket f(a, b) \rrbracket$ and $\llbracket g(a, b) \rrbracket$. $\mathcal{V}$ outputs $f(a, b) \stackrel{?}{=} g(a, b)$.

Figure 16: Protocol for checking set-inclusion.

$(a_0, a_1, a_2) \mapsto a_0 + \alpha a_1 + \alpha^2 a_2$ with $\alpha \in \mathbb{F}_{\text{ZK}}$. For any fixed bad triple this occurs with probability at most $2/|\mathbb{F}_{\text{ZK}}|$, and a union bound over N samples gives at most $2N/|\mathbb{F}_{\text{ZK}}|$. Thus, except with probability $\text{negl}(\kappa) + 2N/|\mathbb{F}_{\text{ZK}}|$, acceptance implies that each committed S_i is a table row consistent with U_i , and therefore $S_{i,2} = g_T(U_i)$ for all i . Since $\{U_i\}$ are uniform from $\mathcal{F}_{\text{random}}$, the accepted outputs are distributed as $\widehat{\mathcal{D}'}^N$ (and are δ_2 -close to $(\mathcal{D}')^N$ with Lemma 2), as required by $\mathcal{F}_{\text{sample}}$.

Case 3 (Corrupted verifier / zero-knowledge). A corrupted verifier's view consists only of commitments and the $\mathcal{F}_{\text{ZK}}$ transcripts for the asserted relations, and it never receives openings nor the sample values. By the zero-knowledge property of $\mathcal{F}_{\text{ZK}}$, there exists a PPT simulator $\mathcal{S}$ that, on input the public values (including T and N) and the verifier's randomness (including α), produces a transcript computationally indistinguishable from the real view.

Combining the three cases, Π_{sample} realizes $\mathcal{F}_{\text{sample}}$ in the $(\mathcal{F}_{\text{ZK}}, \mathcal{F}_{\text{random}})$ hybrid model (with small deviation probability δ_2 from ideal $\mathcal{D}'$ when sampling uses n -bit fixed-point CDF probabilities according to Lemma 2). $\square$

C Proof of Theorem 2

Proof. According to Definition 4, $w_k = \sum_{r=0}^{m-1} p(|k|m + r)$. Except for the 0-th bucket, each bucket has m elements. For convenience, we evenly split $p(0) = p(0^+) + p(0^-)$, and w_0 is split into w_{0+} and w_{0-} accordingly. So that all the buckets contain m elements. It is clear to see that $\sum_{x \in \mathbb{Z}} p_s(x) = \sum_{x \in \mathbb{Z}} p(x) = 1$.

By Definition 3, we aim to bound:

$$L_s(t) = \sum_{x \in \mathbb{Z}} p_s(x)^\alpha p_s(x + tm)^{1-\alpha}$$

And similarly, for the original discrete DP noise $\mathcal{D}$, we have:

$$L(t) = \sum_{x \in \mathbb{Z}} p(x)^\alpha p(x + tm)^{1-\alpha}$$

Now we prove that $L_s(t) \leq L(t)$ for any $t \in \mathbb{Z}$ and $\alpha > 1$.

$$\begin{aligned} L_s(t) &= \sum_{k \in \mathbb{Z}} \sum_{r=0}^{m-1} p_s(km + r)^\alpha p_s(km + r + tm)^{1-\alpha} \\ &= \sum_{k \in \mathbb{Z}} \sum_{r=0}^{m-1} \frac{w_k^\alpha}{(m)^\alpha} \frac{w_{k+t}^{1-\alpha}}{(m)^{1-\alpha}} \\ &= \sum_{k \in \mathbb{Z}} w_k^\alpha w_{k+t}^{1-\alpha} \end{aligned}$$

For $x = km + r$, let $p(x) = p(km + r) = w_k a_{k,r}$, where $a_{k,r} \in [0, 1]$. Since $\sum_{r=0}^{m-1} p(km + r) = w_k$, we have $\sum_{r=0}^{m-1} a_{k,r} = 1$. Then for the original distribution, we have:

$$\begin{aligned} L(t) &= \sum_{k \in \mathbb{Z}} \sum_{r=0}^{m-1} p(km + r)^\alpha p(km + r + tm)^{1-\alpha} \\ &= \sum_{k \in \mathbb{Z}} \sum_{r=0}^{m-1} w_k^\alpha a_{k,r}^\alpha w_{k+t}^{1-\alpha} a_{k+t,r}^{1-\alpha} \\ &= \sum_{k \in \mathbb{Z}} \left(w_k^\alpha w_{k+t}^{1-\alpha} \sum_{r=0}^{m-1} a_{k,r}^\alpha a_{k+t,r}^{1-\alpha} \right) \end{aligned}$$

Let $c_r = \frac{a_{k,r}}{a_{k+t,r}}$, so $\sum_{r=0}^{m-1} c_r a_{k+t,r} = 1$. By Jensen's inequality:

$$\sum_{r=0}^{m-1} a_{k,r}^\alpha a_{k+t,r}^{1-\alpha} = \sum_{r=0}^{m-1} c_r^\alpha a_{k+t,r} \geq \left(\sum_{r=0}^{m-1} c_r a_{k+t,r} \right)^\alpha = 1$$

Thus we have:

$$L(t) \geq \sum_{k \in \mathbb{Z}} w_k^\alpha w_{k+t}^{1-\alpha} = L_s(t), \quad \forall t \in \mathbb{Z}$$

Thus, the Rényi divergence of the staircase noise is upper bounded by the original discrete DP noise:

$$D_\alpha(\mathcal{D}_s || \mathcal{D}_s + \Delta) = \frac{1}{\alpha - 1} \log(L_s(t)) \leq \frac{1}{\alpha - 1} \log(L(t)) \leq \frac{1}{2} \epsilon^2 \alpha$$

So far, we have proved that the Rényi divergence of staircase DP noise is upper bounded by the original discrete DP noise when $\Delta = tm, t \in \mathbb{Z}$. In the following, we prove that the divergence increases with the increase of $|\Delta|$. That is, if we pick m carefully such that the sensitivity Δ is multiple of m , the maximum Rényi divergence of staircase noise is upper bounded by the original DP noise. To prove this, it is equivalent to prove that $L_s(t)$ increases monotonically as t increases when $t > 0$, and decreases monotonically as t increases when $t < 0$. Without loss of generality, we consider the case of $t > 0$, and the proof can be generalized to $t < 0$.

Consider integer-valued random variable R with pmf $\Pr[R = k] = \frac{w_k^\alpha}{Z}, k \in \mathbb{Z}, Z = \sum_{j \in \mathbb{Z}} w_j^\alpha \leq 1$. Define a convex and strictly decreasing function $g(x) = x^{1-\alpha}$ ($x > 0$). Then:

$$L_s(t) = Z \mathbb{E}[g(w_{R+t})], (t \in \mathbb{Z}_{>0})$$

Let $Y_t := w_{R+t}$. Y_t is stochastically smaller when t is larger:

$$Y_{t+1} \preceq_{st} Y_t \iff \Pr[Y_{t+1} \geq y] \leq \Pr[Y_t \geq y], \forall y \in [0, 1], t \in \mathbb{Z}_{>0} \quad (1)$$

Since w_k is unimodal and maximizes at w_0 , and we have $w_{\pm i} \geq w_{\pm(i+1)}, \forall i \in \mathbb{Z}_{>0}$. For $t > 0$, a further shift on t will not increase the probability $\Pr[Y_t \geq y]$. Thus Eq. 1 holds.

Take $h(x) = -g'(x) \geq 0, \forall x > 0$. From integration by parts, we have:

$$\mathbb{E}[g(Y_t)] = \lim_{\epsilon \rightarrow 0^+} g(\epsilon) - \int_{\epsilon}^1 h(u) \Pr[Y_t > u] du$$

Similarly:

$$\mathbb{E}[g(Y_{t+1})] = \lim_{\epsilon \rightarrow 0^+} g(\epsilon) - \int_{\epsilon}^1 h(u) \Pr[Y_{t+1} > u] du$$

Taking the subtraction, we have:

$$\begin{aligned} & \mathbb{E}[g(Y_{t+1})] - \mathbb{E}[g(Y_t)] \\ &= \lim_{\epsilon \rightarrow 0^+} \int_{\epsilon}^1 h(u) (\Pr[Y_t > u] - \Pr[Y_{t+1} > u]) du \end{aligned}$$

Since $h(u) \geq 0$ and $\Pr[Y_t > u] - \Pr[Y_{t+1} > u] \geq 0$, we have:

$$\mathbb{E}[g(Y_{t+1})] \geq \mathbb{E}[g(Y_t)]$$

Thus:

$$L_s(t+1) \geq L_s(t)$$

Thus $L_s(t)$ is a monotonically increasing function and it is maximized when $t = \frac{\Delta}{m}$. $\square$

D Multivariate Staircase DP Noise

Our analysis in § 5.1 applies to single-dimensional cases or multidimensional scenarios with individual sensitivity bounds for each dimension (e.g., the aggregation of user's feature vectors like age, income, etc., each feature dimension has their own sensitivity bound). However, there exist applications that need to bound the L_p norm of the vector (e.g., in DPSGD, we bound the L_2 norm of the gradients if we use Gaussian mechanism or bound L_1 norm if we use Laplace mechanism.). Therefore, we need a more general analysis that do not rely on bounding m using the sensitivity Δ , because the sensitivities for each dimension can now be arbitrary as long as their L_p norm is bounded by Δ . Thus, we cannot select m to guarantee that $\Delta_i = km$ for each dimension i .

Theorem 4 (Privacy Guarantee of Multivariate Discrete Staircase DP Noise). *The multivariate staircase noise distribution $\mathcal{D}_s$ satisfies $\frac{1}{2}(\beta\epsilon)^2$ -CDP, as long as the original DP discrete noise satisfies $\frac{1}{2}\epsilon^2$ -CDP under L_p norm sensitivity bound Δ_p , and its PMF is symmetric and monotonous, i.e., $p(x) = p(-x)$ and $p(x) \geq p(x+1)$, $\forall x \in \mathbb{Z}_{\geq 0}$. β is a bounded constant.*

Proof. Let $\beta^2 = \max_{|t| \leq \Delta, t \in \mathbb{Z}} \frac{D_\alpha(\mathcal{D}_s || \mathcal{D}_s + t)}{D_\alpha(\mathcal{D} || \mathcal{D} + t)}$. Then for any $|t| \leq \Delta_p$:

$$D_\alpha(\mathcal{D}_s || \mathcal{D}_s + t) \leq \beta^2 D_\alpha(\mathcal{D} || \mathcal{D} + t) \leq \frac{1}{2}(\beta\epsilon)^2$$

Consider $|t| = km + r, k \in \mathbb{Z}_{\geq 0}, r \in [m]$, in § 5.1, we have proven that the Rényi divergence of staircase noise increases with the increase of $|t|$, thus:

$$D_\alpha(\mathcal{D}_s || \mathcal{D}_s + t) \leq D_\alpha(\mathcal{D}_s || \mathcal{D}_s + t'), |t'| = (k+1)m$$

For $|t'| = (k+1)m$, we have:

$$D_\alpha(\mathcal{D}_s || \mathcal{D}_s + t') \leq D_\alpha(\mathcal{D} || \mathcal{D} + t')$$

Thus:

$$\beta^2 \leq \frac{D_\alpha(\mathcal{D}_s || \mathcal{D}_s + t)}{D_\alpha(\mathcal{D} || \mathcal{D} + t)} \leq \frac{D_\alpha(\mathcal{D} || \mathcal{D} + t')}{D_\alpha(\mathcal{D} || \mathcal{D} + t)} \quad (2)$$

We can easily upper bound the RHS of Eq. 2 for the original DP noise. For instance, if $\mathcal{D}$ is discrete Gaussian, $\frac{D_\alpha(\mathcal{D} || \mathcal{D} + t')}{D_\alpha(\mathcal{D} || \mathcal{D} + t)} \leq (1 + \frac{1}{k})^2$, and for discrete Laplace $\frac{D_\alpha(\mathcal{D} || \mathcal{D} + t')}{D_\alpha(\mathcal{D} || \mathcal{D} + t)} \leq 1 + \frac{1}{k}$, where $k = \lceil \frac{\Delta_p}{m} \rceil$. Thus β is a bounded constant. $\square$

In Theorem 4, we do not require the sensitivity to be multiple of bucket size ($\Delta = km$). Analytically, we can upper bound β for different noise distributions as we rigorously prove above, but such bound can be loose when the sensitivity is small. In practice, we recommend computing the actual value of β by computing the Rényi divergence for possible discrete sensitivity values, which will be much tighter.

E Proof of Theorem 3

In Appendix D, we have shown how to bound the privacy loss for discrete staircase DP noise. However, the same technique cannot be directly applied to the continuous staircase DP noise, as the sensitivity of each dimension can be infinitely small, making the privacy loss ratio compared to original continuous DP distribution infinite. Below, we give the rigorous proof on bounding the privacy loss using an alternative approach.

Proof. Consider that the original continuous DP noise satisfies $\frac{1}{2}\epsilon^2$ -CDP. For the multivariate case, the Rényi divergence sum of all d dimensions are bounded:

$$\sum_{i=1}^d D_\alpha(\mathcal{D} || \mathcal{D} + t_i) \leq \frac{1}{2}\epsilon^2 \alpha, \sum_{i=1}^d t_i^p \leq \Delta_p^p$$

If all the t_i satisfy that $t_i = k_i m, k_i \in \mathbb{Z}$ and m is the bucket size, then by generalizing Theorem 2 to the continuous case, we have:

$$\sum_{i=1}^d D_\alpha(\mathcal{D}_s || \mathcal{D}_s + t_i) \leq \frac{1}{2}\epsilon^2 \alpha, \sum_{i=1}^d t_i^p \leq \Delta_p^p, t_i = k_i m, k_i \in \mathbb{Z}$$

Protocol $\Pi_{\text{TConstruct}}$

Input: Additive and truncated DP noise distribution $\mathcal{D}'$, number of samples N .

Protocol:

1. Compute the PMF $p_i = \mathcal{D}'(\mathbf{x}_i), i \in [0, M]$ of the noise distribution $\mathcal{D}'$.
2. Calibrate the PMF for symmetric distributions: $p_i = 2p_i, i \in [1, M]$ and normalize the PMF: $p_i = p_i / \sum_{j=0}^M p_j$.
3. Compute the CDF table $p_i^{\max} = \sum_{j=0}^i p_j, p_i^{\min} = p_i^{\max} - p_i, i \in [0, M]$.
4. Quantize the CDF probabilities to fixed-point using n bits according to Lemma 2.

Output: Public table $T = \{p_i^{\min}, p_i^{\max}, \mathbf{x}_i\}_{i \in [M]}$ for noise sampling.

Figure 17: Protocol to construct the public CDF table.

However, we cannot guarantee that all the t_i satisfy the above condition. Thus, we enforce this condition by spending slightly more privacy budget.

In Appendix C, we prove that the Rényi divergence of the staircase noise is monotonous to the sensitivity t_i . Thus, we upper bound the Rényi divergence of the continuous staircase DP noise by increasing t_i to t'_i such that $t'_i = k_i m, k_i \in \mathbb{Z}$. In the worst case, for each dimension, we increase t_i by m . Thus, we can upper bound the Rényi divergence using the original continuous DP distribution with a new ϵ' . Since ϵ' corresponds to the new increased sensitivity Δ'_p . By Minkowski inequality, we have:

$$\Delta'_p = \|\mathbf{t} + \mathbf{1}m\|_p \leq \|\mathbf{t}\|_p + md^{1/p} \leq \Delta_p + md^{1/p}, p \geq 1$$

By setting $m = c\Delta_p/d^{1/p}$, we get the new $\Delta'_p \leq (1+c)\Delta_p$. Thus we can rigorously bound the privacy loss using the DP distribution with sensitivity $(1+c)\Delta_p$ and obtain the privacy budget bound $\epsilon' = (1+c)^{p/2}\epsilon$. $\square$

F Table Construction.

Firstly, we determine the truncation bound M according to a given additive DP noise distribution $\mathcal{D}$ and Tab. 1, and obtain truncated DP noise distribution $\mathcal{D}'$. Since the distribution is symmetric, we compute the probability mass function (PMF) p_i for values $\mathbf{x}_i, i \in [0, M]$ and double the probabilities for $i \in [1, M]$. After normalization, we compute the CDF for the truncated unsigned distribution as $\text{CDF}_i = \sum_{j=0}^i p_j$. Fig. 17 describes the table construction for discrete distributions in detail. When it comes to continuous distributions, the CDF will be the integral of the distribution's PDF, the other parts remain the same.